

Essential Statistics for Researchers

A beginner-to-advanced field guide: read your data, choose the right test, and design studies that hold up

Press Space to begin · o for the slide overview

What you'll learn

One continuous path from "what is a mean?" to "which test, and why," split into eleven short parts. Every part ends with a quick self-check.



1 • Foundations

What statistics is, and its two jobs: estimation and hypothesis testing.



2 • Know your data

Nominal, ordinal, interval, ratio: the scale decides everything downstream.



3 • Distributions

Frequency tables, histograms, box & violin plots, the median, correlation.



4 • Choosing a test

Normality & equal variance, then parametric vs. non-parametric.



5 • Categorical data type

Counts, contingency tables, chi-square, Fisher's, logistic & Poisson: tests for frequencies.



6 • Rank-based categories

Ordered data: Mann-Whitney, Kruskal-Wallis, Spearman, and ordinal proportional-odds / probit regression.



7 • Hypothesis testing

Null & alternative, α , tails, p-values, and Type I / Type II error.



8 • Beyond p-values

Effect size, statistical power, sample size, and confidence intervals.



9 • Study design

PICOT/PECOT, accuracy vs. precision, and the hierarchy of evidence.



10 • Tools & reporting

Vector vs. raster figures, and the software that does the work.



11 • Cheat-sheet

A which-test decision table, glossary, and take-home messages.



TIP

This whole guide is one Slidew deck. Press  at any time to see every slide at once and jump around. It doubles as a reference you can return to.

■ PART 1 · FOUNDATIONS

What statistics actually does

Before any test or p-value: what is statistics *for*, and what are its two core jobs?

What is statistics?

Statistics is a quantitative understanding of a phenomenon, the thing you measure or observe. You can't watch the whole world, so you *sample* it, then apply methods that turn those numbers into understanding.



Collect

Survey, measure, or test your subjects to gather raw data.



Analyze

Apply methods: compare means with a t-test or ANOVA, model relationships.



Communicate

Report it: figures, tables, p-values, degrees of freedom, sample sizes.



THE REAL GOAL

To gain an **objective understanding of data that contains variance**. Variability is not noise to hide, but the very thing statistics is built to handle.

The two jobs of statistics

Almost everything in this deck serves one of these two purposes. Keep them in mind, because they shape which numbers you report.



1 • Estimation & inference

Use a **partial** dataset (your sample) to make statements about an entire **population** you could never measure in full.



2 • Hypothesis testing

Decide whether to **reject** or **fail to reject** a stated hypothesis, based on what the data show.

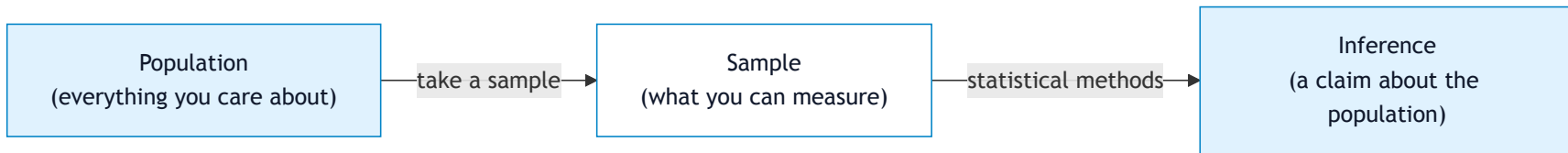


TIP

A bonus you get for free: a good analysis also **reveals the underlying structure** of your data, letting you summarize and classify it, not just test it.

Why we sample

You are one researcher (or one team). You almost never get to measure the whole **population**, so you measure a **sample** and reason back to the whole.



The leap from **sample** to **population** is exactly where uncertainty enters, and why we need every tool in the rest of this deck.

The vocabulary you'll keep seeing

The setup

Population: everything you want to conclude about.

Sample: the subset you actually measure.

Variance: how spread out the data are.

Parameter: a true value in the population.

The decision

Hypothesis: a specific claim about the population that the data can support or reject.

p-value: if there were truly no effect, the chance of a result this strong *or stronger* by luck alone.

α (alpha): if the p-value falls below it, you call the result significant; chosen in advance.

The report

Degrees of freedom: how many values can still vary once a summary like the mean is fixed.

Effect size: the difference's size relative to the data's spread (Cohen's d), capturing whether it *matters*, not just that it exists.

CI: confidence interval, a plausible range.

Self-check

Your study measures grip strength in 40 patients to say something about *all* patients with a condition. **Which of the two jobs is that, and what is the population vs. the sample?**



ANSWER

That's **estimation / inference**, using a sample to make a claim about a population.

- **Population:** all patients with the condition.
- **Sample:** the 40 patients you measured.

The variability *between* those 40 is what tells you how much to trust the leap to "all patients."

• PART 2 · KNOW YOUR DATA

The four scales of measurement

The single most important question before any analysis: **what kind of data is this?** The scale decides which calculations even make sense.

Two big families

Every variable is either a **label** or a **number you can do arithmetic on**. Get this split right and the rest follows.



Qualitative · categorical

Data sorted into **categories**. Covers the **nominal** and **ordinal** scales. Usually analyzed with **non-parametric** statistics.



Quantitative · continuous

Measured **amounts**. Covers the **interval** and **ratio** scales. The home of **parametric** statistics (means, ANOVA).



TIP

Four scales, in increasing order of "how much math is allowed":

nominal

ordinal

interval

ratio

Nominal scale

Categories with no order. The values are just *names*. Even if you code them as numbers, the numbers are **dummy variables** whose size means nothing.

- Example: sex (male / female), coded 1 / 2
- Example: blood type, treatment group, species

There is no "bigger" or "smaller": 2 is not more than 1, just *a different label*.



WHAT YOU MAY NOT DO

Calculating an **average** is meaningless. The "mean sex" of a sample is a number with no interpretation.

What you analyze instead

- Counts and proportions
- The **mode** (most frequent category)
- A **chi-square** test for association

Ordinal scale

Categories with a rank, but uneven gaps. Numbers show *order only*, not how far apart things are.

- Example: journal rank (1st, 2nd, 3rd quartile)
- Example: pain rated mild / moderate / severe
- Example: Likert "disagree → agree"

You know 3rd beats 2nd, but not by *how much*: the distance between ranks isn't fixed.



WHAT YOU MAY NOT DO

Means and standard deviations are still **meaningless**, because only the relative ranks carry information.

What you analyze instead

- The **median** and **quartiles**
- **Wilcoxon–Mann–Whitney** (two groups)
- **Kruskal–Wallis** (more groups)

These are **non-parametric** rank tests.

Interval scale

Equal, meaningful gaps, but no true zero. Differences are comparable; the zero point is *arbitrary*, so ratios are not.

- Example: temperature in °C
- Example: IQ score

20 °C is as far from 30 °C as 30 °C is from 40 °C (equal intervals), but 40 °C is **not** "twice as hot" as 20 °C, because 0 °C isn't a true absence of heat.



NOW WE'RE PARAMETRIC

With equal intervals you can compute **averages and totals**, so means-based tests like the **t-test** and **ANOVA** become valid.

Ratios ("twice as much") are the one thing still off-limits, because that needs a true zero.

Ratio scale

Equal gaps *and* a true zero. Zero means "none," so even ratios are meaningful.

- Example: length, weight
- Example: blood pressure, grip strength

A 20 kg sample really is **twice** 10 kg, because 0 kg means no mass at all.



EVERYTHING IS ALLOWED

Averages, totals, **and** ratios: the full parametric toolkit applies.



For choosing a test, interval vs. ratio rarely matters, since both are continuous and both support means. The big divide is **categorical vs. continuous**.

The four scales at a glance

Nominal

Labels, no order

Sex, blood type

✗ mean ✗ rank

Counts · mode · χ^2

Ordinal

Ranked, uneven gaps

*Pain level, journal
rank*

✗ mean ✓ rank

Median · rank tests

Interval

Equal gaps, no true 0

°C, IQ

✓ mean ✗ ratio

t-test · ANOVA

Ratio

Equal gaps, true 0

Weight, BP

✓ mean ✓ ratio

t-test · ANOVA

Why this is the first decision

Categorical data → **non-parametric** tests.

Continuous data → **parametric** tests (means, ANOVA) if assumptions hold.



TIP

Naming the scale of each variable tells you, before you touch any software, which half of the statistics world you're working in.

Self-check: classify these

For each variable, name the **scale**, and whether means are allowed.

- **Resting heart rate (bpm)** → *ratio*: true zero, means ✓
- **Tumor stage I / II / III / IV** → *ordinal*: ranked, means ✗ (use the median)
- **Eye colour** → *nominal*: labels only, means ✗
- **Temperature of a water bath (°C)** → *interval*: equal gaps, means ✓, but no "twice as hot"

 Trick of the trade: if "zero" means "none of it," you're on a ratio scale.

• PART 3 • DISTRIBUTIONS

Look at your data before you test it

Exploring the *shape* of your data tells you which tests are valid, and reviewers increasingly demand to see it.

Explore first, test later

Modern journals want to see **variability and distribution**, not just a mean with an error bar. Before any test, summarize your data with tables and graphs.



Why it matters to you

The shape of your data decides whether a mean is even meaningful, and which test is legitimate.



Why reviewers want it

A box or violin plot shows the **full spread**; a bare mean $\pm$ SE hides it.



TIP

The cheapest first step after collecting data: a **frequency distribution table**. It costs nothing and immediately reveals the shape.

Frequency distribution table

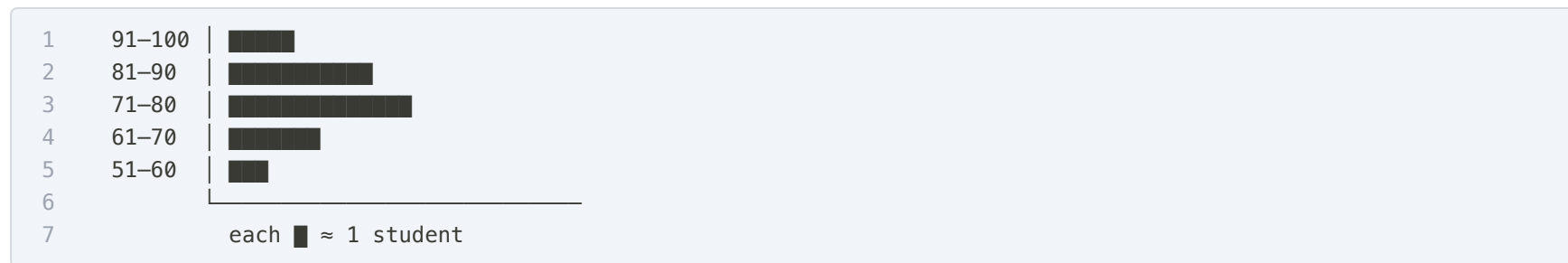
Group raw values into **levels** (bins), then count how many land in each. The **cumulative ratio** always climbs to 100%.

Score range	Frequency (n)	Cumulative n	Ratio	Cumulative ratio
51–60	3	3	7.5%	7.5%
61–70	7	10	17.5%	25.0%
71–80	14	24	35.0%	60.0%
81–90	11	35	27.5%	87.5%
91–100	5	40	12.5%	100%

 40 students total. At a glance: most scores cluster in the 71–80 band, and 60% scored 80 or below.

The same table, as a histogram

Plot the counts and the shape jumps out, here with a **bin width of 10**:



A histogram conveys in one glance what the table lists row by row.

The interval trap

The *same data* tells different stories depending on the **bin width** you pick. This is a real source of accidental spin.

Too narrow (bins of 2)



Spiky and jagged: random wobble looks like real structure.

Just right (bins of 10)



A clean single peak: the actual trend is visible.

Too wide (bins of 40)



Over-smoothed: the shape is flattened into nothing.



CHOOSE THE INTERVAL DELIBERATELY

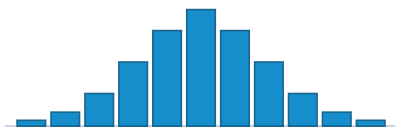
Bin width is a **choice that shapes the story**. Try a few and pick the one that honestly shows the trend, not the prettiest one.

Common histogram shapes

Recognizing the shape tells you whether a mean and a normality-assuming test make sense.



Normal (bell)



Symmetric. Mean $\approx$ median. Parametric tests are happy.



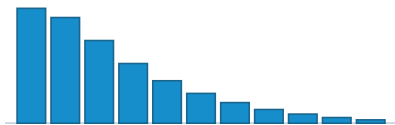
Bimodal / multimodal



Two peaks: often two subgroups mixed together.



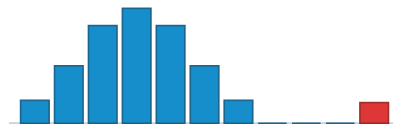
Skewed



A long tail (here, to the right). Mean is dragged toward the tail.



Normal + outlier



Mostly bell-shaped, but one stray value (e.g. 130) distorts the mean.

The median, exactly

The **median** is the middle value: 50% of the data sit above it, 50% below. It's the go-to centre for skewed data and outliers.

Odd number of points

1	88	90	[92]	94	96
2			↑		
3			middle value		

Median = 92 (just take the centre value).

Even number of points

1	88	90	[91	92]	94	96
2			↑	↑		
3			two middle values			

Median = $(91 + 92) / 2 = 91.5$ (average the two middles).

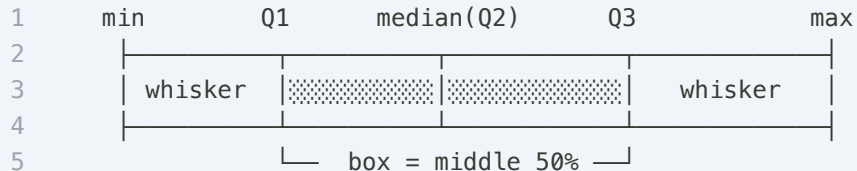


TIP

Because it ignores how *far* outliers are, the median barely moves when one extreme value appears, unlike the mean. That's why it pairs with non-parametric tests.

Box-and-whisker plots

A box plot draws the distribution from its **quartiles**, the values that cut the data into four equal-count slices.



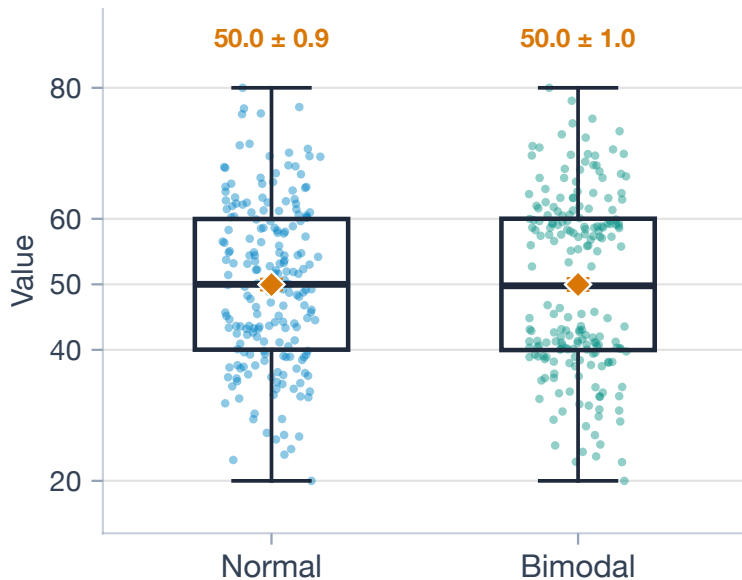
Box = the interquartile range (Q1→Q3), the middle 50% of the data.



Whiskers reach to the min and max, or in the common **Tukey** version to $1.5 \times \text{IQR}$, with outliers drawn as dots.

The box plot's blind spot

A box plot knows only **five numbers** (min, Q1, median, Q3, max), so very different data can produce the *identical* box plot.



Same box plot *and* near-identical mean ($\diamond$) $\pm$ SE, yet different shapes: the box can't see shape.

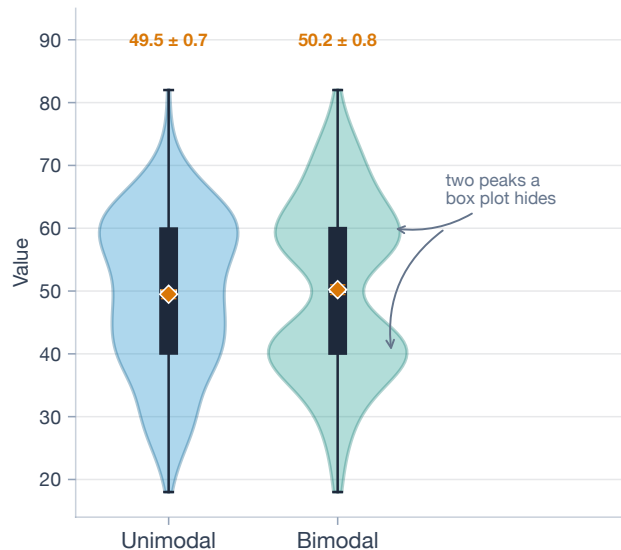


TIP

This is exactly why **violin plots** exist: same five numbers, but the *shape* is revealed (next slide).

Violin plots

A violin plot is a box plot that also shows **probability density**, how likely values are at each level. **Wider = more data piled there.**



Why editors love them

- Shows the **whole shape**, not just five summary numbers.
- Reveals bimodality a box plot would hide.
- A bare **mean $\pm$ SE** throws all of this away.

Identical box plot and mean ($\diamond$) $\pm$ SE, yet only the violin reveals the two peaks.

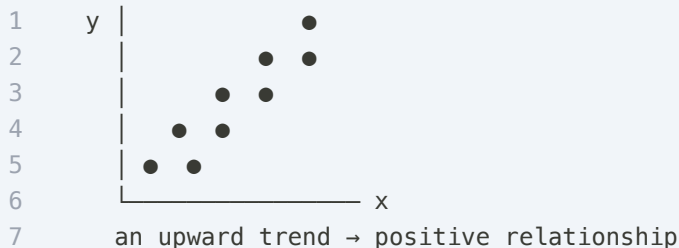


TIP

In one figure, box and violin plots answer the reviewer's real question: "what does your data actually look like?"

Scatterplots & correlation

To explore a relationship between **two** variables, plot one against the other.



What you can measure

- **Pearson's r**: strength & direction of a *linear* relationship, from -1 to +1.
 - **R²**: the fraction of variation the line explains.
 - Fit a **linear regression** line to predict y from x.
- r = **0.94** → a strong positive correlation.



CORRELATION ≠ CAUSATION

A tight correlation says two things move together, **not** that one causes the other. Always interpret **r** alongside its p-value and a healthy dose of skepticism.

Self-check

Your reaction-time data has a long right tail and one extreme value at 130 ms. **Which centre should you report (mean or median), and which plot best shows the shape?**



ANSWER

Report the **median**: the skew and the outlier would drag the mean toward the tail and misrepresent the typical case.

For the figure, a **box-and-whisker** (or **violin**) plot is ideal: it shows the median, the spread, and the outlier all at once, rather than hiding them behind a single error bar.

■ PART 4 · CHOOSING A TEST

Parametric or non-parametric?

The choice isn't a coin flip; your data's distribution decides it. Two assumption checks point the way.

The gateway: check assumptions first

Parametric tests (t-test, ANOVA) **assume** your data are normal with equal variances. Before using one, *test those assumptions*. This is the most important step before picking a test.



1 • Normality

Is the distribution roughly **symmetric about the mean** (mean $\approx$ median)?

Test with **Shapiro–Wilk** (among others).



2 • Homoscedasticity

Do the groups have **equal variances**?

Test with **Bartlett's** or **Levene's** test.



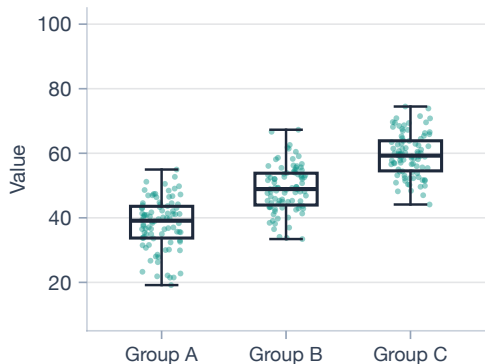
NOTE

A "normal" distribution can be wide or narrow; what matters is that it's **symmetric**, with the mean and median sitting at the same place.

Homoscedasticity: equal spread?

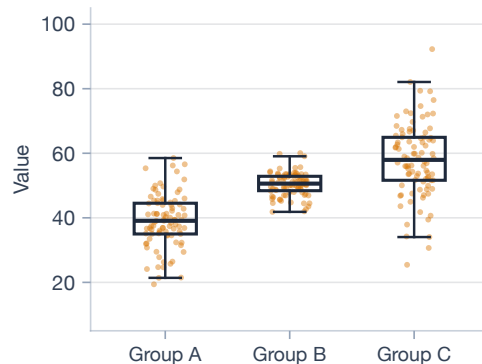
The word just means **the groups vary by the same amount**, so they can be assumed to come from comparable populations.

Equal variances ✓



Same spread *despite different means*: homoscedastic. ANOVA's assumption holds.

Unequal variances ✗



Different spreads: heteroscedastic. The assumption fails.

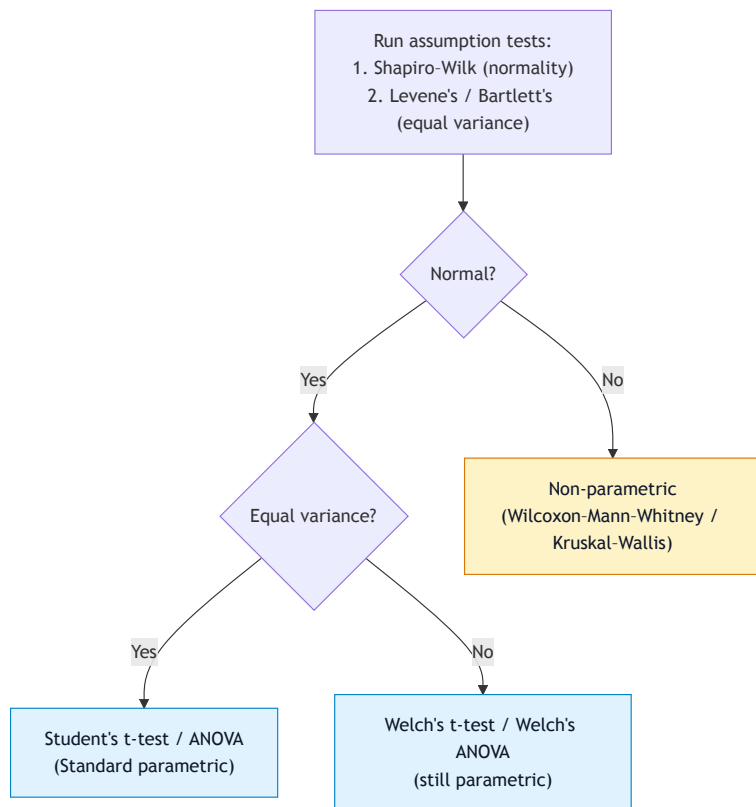


NOTE

A **significant** Bartlett's / Levene's test ($p < 0.05$) means you **reject** equal variances: the spreads really do differ.

The decision rule

Run the assumption tests, then read off the path:



TIP

Failing an assumption isn't a dead end; the fix depends on *which*:

- **Non-normal** data → its non-parametric twin (**Kruskal-Wallis**)
- **Normal** but unequal variance → **Welch's** correction, still parametric

Why continuous data can still be non-parametric

"Continuous → parametric" is a **starting candidate**, not the verdict. The test is settled in **two stages**, and a continuous variable can still land on a non-parametric one.

Stage 1 · Is a mean meaningful?

The **data type** decides:

- **Categorical** → mean meaningless → non-parametric / counts (*branch ends*)
- **Continuous** → mean meaningful → **on to Stage 2** →

Stage 2 · Do the assumptions hold?

For **continuous** data, test **normality**:

- Hold **normality** ✓ → **parametric** ([Welch's] t-test, ANOVA)
- Fail **normality** ✗ → **non-parametric** (Wilcoxon–Mann–Whitney, Kruskal–Wallis)

Continuous data → check assumptions → **pass** → **parametric** · **fail** → **non-parametric**



TIP

Why non-parametric works on continuous data too: these tests rank the values instead of using the raw numbers, dropping the interval detail. That makes them more robust to skew and outliers, but slightly less powerful when the parametric assumptions truly hold.

Parametric ↔ non-parametric: the map

Pick your **question** on the left, then read **across the row** to the parametric test and its non-parametric twin, where one exists.

Your question	Parametric — <i>assumes a distribution</i>	Non-parametric — <i>distribution-free</i>
Compare 2 groups	Independent t-test	Wilcoxon–Mann–Whitney
Compare 3+ groups	One-way ANOVA	Kruskal–Wallis
Measure a relationship	Pearson's r / regression	Spearman's ρ
Test a categorical association	Logistic regression	Chi-square
Model a count	Poisson / NB	—

Don't run many separate t-tests

Comparing several groups pair-by-pair inflates the **false-positive** rate, because each extra test is another chance to flag a difference that isn't really there.

The trap — every pair

Three groups **A, B, C** already need **three** tests:

- A vs B
- A vs C
- B vs C

More comparisons → higher odds that *at least one* looks significant by chance alone.

Better — one overall test first

- **One-way ANOVA**: parametric
- **Kruskal-Wallis**: non-parametric

Both answer a single question:

*Is there evidence that **at least one** group differs?*



NOTE

The overall test tells you **whether** any group differs, not **which** one. Pinpointing the pair is the job of a post-hoc test. →

Which groups differ? Post-hoc tests

A significant ANOVA or Kruskal–Wallis says *some* group differs; a **post-hoc** test then compares specific pairs while **controlling** the false-positive risk of multiple comparisons.



Tukey's test

Use **after one-way ANOVA** (parametric). Compares every pair, holding the overall error rate in check.



Dunn's test

Use **after Kruskal–Wallis** (non-parametric). The rank-based equivalent for pinpointing which pairs differ.



BONFERRONI CORRECTION

A simpler alternative when you *do* run several pairwise tests: split α across them. For **three** comparisons,

$$\alpha_{\text{adjusted}} = \frac{0.05}{3} = 0.0167$$

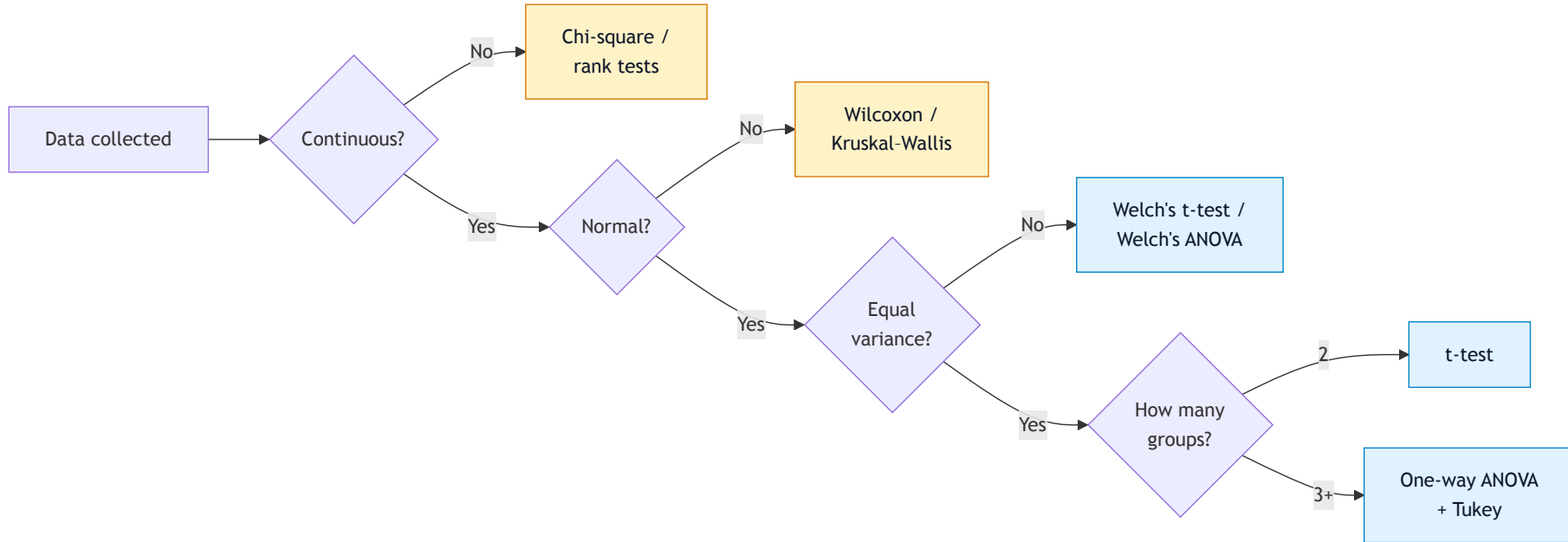
so a result counts as significant only when $p < 0.0167$, stricter than the usual $p < 0.05$.



NOTE

In short: ANOVA / Kruskal–Wallis tells you *whether* any group differs; Tukey / Dunn tell you *which*. Both keep false positives under control.

The whole decision, in one flow



Self-check

You're comparing a biomarker across **four** treatment groups. **Shapiro–Wilk** says the data are roughly normal ($p = 0.40$), but **Levene's test** comes back $p = 0.01$. Which test do you run, and what follows if it's significant?



ANSWER

The data are **normal** but the variances are **unequal**, so don't jump to Kruskal–Wallis. Use **Welch's ANOVA** (parametric; it drops the equal-variance assumption).

If it's significant, follow up with the **Games–Howell** post-hoc test (the variance-robust analog of Tukey) to find *which* groups differ.

*(Only if normality had **also** failed would you switch to Kruskal–Wallis → Dunn's.)*

■ PART 5 · CATEGORICAL DATA TYPE

Counting categories

When there's no mean to take, you count, and a whole family of tests works on the frequencies.

The categorical off-ramp: count the frequencies

Categorical data sorts observations into **labels**: blood type, improved/unchanged, survived/died. There are no numeric values to add up, so an average is meaningless.



No number to average

"The **mean blood type is 2.4**" says nothing, because labels have no magnitude to sum or rank.



So count instead

Tally **how many** observations fall in each category, then analyze those counts, usually in a **contingency table**.

Treatment	Improved	Unchanged
Drug A	34	16
Drug B	22	28

Counts, not averages: **68%** improved on A vs **44%** on B, and a χ^2 test asks if the gap is real.



NOTE

This is why the categorical branch **ends at Stage 1**: with no mean there's no normality check to run, so counts get their own family of

Rank-based vs count-based: is there an order?

Categorical data branches two ways: **rank-based** and **count-based** tests answer **different questions** and need **different data**. The deciding question: *is there a rank order?*



Ordinal → rank-based

Categories with an **order** (mild < moderate < severe). Rank-based tests exploit that order: **Mann–Whitney, Kruskal–Wallis, Spearman**.



Nominal → count-based

Categories with **no order** (blood type A/B/AB/O). Nothing to rank, so test the **frequencies**: **chi-square, Fisher's, proportion tests**.

A different axis: parametric vs non-parametric

Whether a test is *parametric* is a **separate question** from the **data's type**, and every data type carries tools of **both** kinds.

Non-parametric just means a test assumes **no particular distribution** for the data (*no bell curve required*).

data type	non-parametric · <i>no distribution assumed</i>	parametric · <i>assumes a distribution</i>
nominal / counts	chi-square · Fisher's exact	logistic · Poisson · negative-binomial
ordinal	Mann–Whitney · Kruskal–Wallis · Spearman	proportional-odds (ordinal) logistic · probit
interval / ratio (<i>continuous</i>)	Mann–Whitney · Wilcoxon · Kruskal–Wallis · Spearman · bootstrap / permutation	t-test · ANOVA · Pearson · linear regression

 **NOTE**
"Non-parametric" is a **column**, not a **row**: every data type has one. The parametric side asks for more: an **assumed distribution**.

Continuous data: parametric or non-parametric?

Interval / ratio data takes **either** family; the split is whether the test **assumes a distribution**.



Parametric — assumes a distribution

t-test · ANOVA · Pearson correlation · linear / OLS regression.

These assume an underlying distribution, roughly **normal**, often with **equal variance**.



Non-parametric — assumes none

Mann–Whitney · Wilcoxon signed-rank · Kruskal–Wallis · Spearman · Kendall's τ · sign test, plus resampling (bootstrap, permutation).

They convert the values to **ranks** (or **resample**), so the **normality assumption drops away**.



WHEN TO ACTUALLY GO NON-PARAMETRIC

Skewed data, outliers, tiny samples, or unequal variances → fall back from the t-test / ANOVA to **Mann–Whitney / Kruskal–Wallis** (or a permutation test). When the **assumptions hold**, use the **parametric** version, which has **more power**.

Rank-based vs frequency-based

Both handle categorical data but pull opposite ways: *ranks* exploit order, *counts* just tally.

	Rank-based — ordinal	Frequency-based — counts
Data it needs	ordered values (<i>ordinal</i>), or non-normal continuous	<i>unordered (nominal)</i> categories
Works on	the rank of each observation	the tally in each cell
Question	does one group rank <i>higher / lower</i> , or do the <i>ranks</i> correlate?	are the variables <i>associated?</i> (or <i>differ from expected</i>)
Typical tests	Mann–Whitney · Kruskal–Wallis · Spearman	chi-square · Fisher's exact · proportion
Example	pain mild < moderate < severe across drugs	blood type vs disease (yes / no)

 **NOTE** Same categorical branch, opposite tools: **rank tests need an order**; **count tests need none** and only ask *how many* fall in each cell.

Same trial, reshaped for each test

Only **chi-square** uses the 2×2 table as-is. The other three reshape the same Drug A vs Drug B data: a **smaller sample**, a **single drug**, or a **count** instead of a label.

Test	The data you use	Example
χ^2 independence	the full 2×2 table, large n	A 34/16, B 22/28 $\rightarrow$ 68% vs 44%
Fisher's exact	the same 2×2 , but a tiny sample	A 4/1, B 1/4 ($n = 10$)
Goodness-of-fit / proportion	one drug vs a fixed benchmark	Drug A 34/50 = 68% vs $p_0 = 50\%$
Poisson / neg-binomial	a count outcome, not a label	flare-ups per patient-month



NOTE

Flare-up: a temporary worsening or recurrence of symptoms in a chronic disease (e.g. an asthma attack or an arthritis joint flare). Because it can **recur**, it's a *count*, not a one-time yes/no.

One trial, four questions

Same **Drug A vs Drug B** trial: each **question** reshapes the data and points to a different test.



Association

Is improvement **linked to the drug**? → χ^2 test
(Fisher's exact if the trial is tiny)



Benchmark

Does Drug A's **68%** **beat a known 50%**? → **one-proportion test** *(one drug vs a fixed value)*



Odds

Do the **odds of improving** differ by drug? → **logistic regression** *(a binary outcome)*



Rate

Do **flare-ups per patient-month** differ by drug? → **Poisson / NB** *(a count outcome)*

Four tests, four kinds of result

All four tests share the same **null hypothesis** of *no effect* (for the two models, that means a **ratio of 1**), yet each reports a **different kind of result**, all computed from the same Drug A vs Drug B trial.

Test	Null hypothesis (H_0) — "no effect"	What it reports
χ^2 independence	drug and improvement are unrelated	χ^2 , p + Cramér's V (<i>how strong</i>)
One-proportion	Drug A's rate is exactly 50%	a proportion + 95% CI
Logistic regression	drug doesn't shift the odds (OR = 1)	an odds ratio (A vs B) + CI
Poisson / NB	drug doesn't shift the rate (rate ratio = 1)	a rate ratio (A vs B) + CI

Risk	Odds	Rate
risk = events ÷ total people	odds = events ÷ non-events = $p / (1 - p)$	rate = events ÷ person-time
risk ratio = A's risk ÷ B's risk	odds ratio = A's odds ÷ B's odds	rate ratio = A's rate ÷ B's rate

A **ratio of 1** means **no difference** between the two drugs; the further from 1, the larger the effect.

Risk vs odds: two readings of yes/no data

Drug A: 34 of 50 patients improved, 16 did not. Two ways to say "how often."



Risk — share of everyone

events ÷ all people, the everyday *chance*.

$$34/50 = 0.68 = 68\%$$

Counts each person **once** (yes/no) · range $0 \rightarrow 1$ · the intuitive one.



Odds — yes-to-no

events ÷ non-events, the *gambling* sense.

$$34/16 = 2.125 \rightarrow 2.1 \text{ improvers per non-improver}$$

Same data, re-expressed · range $0 \rightarrow \infty$ · no ceiling.



NOTE

The link: $\text{odds} = \text{risk} \div (1 - \text{risk}) = 0.68 \div 0.32 = 2.125$. A **rare** outcome $\rightarrow$ odds $\approx$ risk; a **common** one $\rightarrow$ odds run much larger. Odds have no ceiling, which is why **logistic regression** models them.

Risk ratio and odds ratio

Two ways to compare Drug A with Drug B, both read straight off the trial's 2×2 counts.

	Improved	Unchanged	Total
Drug A	34	16	50
Drug B	22	28	50

Risk ratio — from probabilities

Risk of improving = improved ÷ total:

- Drug A: $34/50 = 0.68$
- Drug B: $22/50 = 0.44$

Risk ratio = $0.68/0.44 \approx 1.55$

*1.55× as **likely** to improve on Drug **A** as on Drug **B**.*

Odds ratio — from odds

Odds of improving = improved ÷ unchanged = $\frac{p}{1-p}$:

- Drug A: $34/16 = 0.68/0.32 = 2.125$
- Drug B: $22/28 = 0.44/0.56 = 0.786$

Odds ratio = $2.125/0.786 \approx 2.70$

*2.7× higher **odds** of improving on Drug **A** than Drug **B**.*

"Risk" just means the **chance** of an outcome, even a good one, like improving.

It only sounds negative because the field grew up studying disease. The risk ratio is also called *relative risk*.

Why a common outcome inflates the odds ratio

Odds and probability are linked by $\text{odds} = \frac{p}{1-p}$. When p is **small**, $\text{odds} \approx p$, so the odds ratio $\approx$ the risk ratio. When p is **large**, the odds balloon, and the odds ratio with them.

Common — our trial (68% & 44%)

	p	odds
A	0.68	2.125
B	0.44	0.786

Risk ratio = $0.68/0.44 \approx 1.55$

Odds ratio = $2.125/0.786 \approx 2.70$

Rare — same risk ratio (6.8% & 4.4%)

	p	odds
A	0.068	0.073
B	0.044	0.046

Risk ratio ≈ 1.55 (unchanged)

Odds ratio = $0.073/0.046 \approx 1.59 \approx$ risk ratio

NOTE

So 2.7 isn't "2.7× more likely to improve": that's the 1.55 risk ratio. The odds ratio only runs higher because improving is *common* here. If the outcome is rare, it collapses back onto the risk ratio.

...and rate: when events repeat over time

Rate = events ÷ **person-time**. Different in kind: it counts events that can happen **more than once**, over follow-up time.

E.g. 30 flare-ups over 300 patient-months ($50 \text{ patients} \times 6 \text{ mo}$) = $\frac{30}{300} = 0.1$ per patient-month is what **Poisson / NB** models.

	Risk	Odds (winners per loser)	Rate
Question	out of everyone, what fraction?	how many yes per <i>no</i> ?	how often per unit time?
Definition	events ÷ <i>all people</i>	events ÷ <i>non-events</i>	events ÷ <i>person-time</i>
Outcome	one-time yes/no	one-time yes/no	recurring count
A person is counted	once	once	possibly many times
Range	$0 \rightarrow 1$	$0 \rightarrow \infty$	$0 \rightarrow \infty$
Drug A example	$34/50 = 0.68$	$34/16 = 2.125$	$0.1 / \text{patient-month}$
Its ratio (A ÷ B)	risk ratio	odds ratio	rate ratio



TIP

risk and **odds** are readings of the *same yes/no snapshot*. Rolling a six on a die: the *risk* is 1/6, but the *odds* are 1/5 (1 six for every 5 non-sixes).

rate counts *repeated events as time passes*. It tallies *events*, not *people*. The only one that needs a time denominator.

Chi-square test of independence

Are two categorical variables associated? In our trial, is **improvement** independent of the drug?

H_0 : both drugs improve at the same rate. Cross-tabulate and test it.

1 · Observed counts O

	Improved	Unchanged	Total
Drug A	34	16	50
Drug B	22	28	50
Total	56	44	100

2 · Expected if independent E

Each cell = $\frac{\text{row} \times \text{col}}{100}$. A-improved = $\frac{50 \times 56}{100} = 28$.

	Improved	Unchanged
Drug A	28	22
Drug B	28	22

3 · Sum the standardized gaps

$$\chi^2 = \sum \frac{(O - E)^2}{E}, \quad \text{df} = (r - 1)(c - 1)$$

Every cell is off by exactly 6 (e.g. 34 vs 28), so:

$$\chi^2 = \frac{6^2}{28} + \frac{6^2}{22} + \frac{6^2}{28} + \frac{6^2}{22} \approx 5.84$$

With $\text{df} = (2 - 1)(2 - 1) = 1$, that clears the **3.84** cutoff ($p \approx 0.016$)

→ **reject independence**: improvement is linked to the drug.



CAUTIONS

Needs **independent** observations and **expected counts** ≥ 5 in ~80% of cells (all ≥ 1); here every E is 22–28. ✓ (Some texts add **Yates' correction** for a 2×2 ; here $\chi^2 \approx 4.9$, same verdict.)



TIP

Tells you *that* there's an association, **not its direction or size**. Follow up with **Cramér's V** (effect size) and **adjusted residuals** (which cells drive it).

Chi-square follow-up: how strong, and where?

The χ^2 on the last slide flagged an association, but *not* its size or direction. Two standard follow-ups finish the story: **quantify** it, then **locate** it.

How strong? — Cramér's V

Rescales χ^2 to a **0–1 effect size** (here $k = \min(r, c) = 2$):

$$V = \sqrt{\frac{\chi^2}{n(k-1)}} = \sqrt{\frac{5.84}{100 \times 1}} \approx 0.24$$

By Cohen's benchmarks (**0.1 small · 0.3 medium · 0.5 large**), this is a **small-to-medium** link: real, but modest.

NOTE

For a 2×2 , V equals the **phi** coefficient, the same story the **risk ratio 1.55 / odds ratio 2.70** tell in clinical terms.



THE COMPLETE VERDICT

χ^2 (5.84, $p \approx 0.016$) → drug and improvement **are associated** · Cramér's $V \approx 0.24$ → the link is **modest** · residuals → **Drug A drives improvement** (B drives "unchanged"). *That* there's an effect, *how big*, and *which way*.

Which cells? — adjusted residuals

Each cell's **adjusted residual**, scaled to $\approx N(0, 1)$, so compare to ± 1.96 :

$$r = \frac{O - E}{\sqrt{E(1 - \frac{\text{row}}{n})(1 - \frac{\text{col}}{n})}}$$

	Improved	Unchanged
Drug A	+2.42	-2.42
Drug B	-2.42	+2.42

Every $|r| > 1.96$, and the + cells sit *above* expected → improvement clusters with **Drug A**.

The plain Pearson residual $(O - E)/\sqrt{E}$ would be only 1.13 here (too small to flag), so the *adjusted* form is what earns the ± 1.96 comparison. In any 2×2 all four residuals share one magnitude, so "which cell" is symmetric here; the breakdown earns its keep in larger $r \times c$ tables, pinpointing the few cells behind a significant χ^2 .

Fisher's exact test — when cells are sparse

Chi-square is a **large-sample approximation**; below ~5 expected per cell it breaks. So picture an **early pilot** of the same trial: just **12 patients**, unevenly split:

The pilot — same trial, tiny n

	Improved	Unchanged	Total
Drug A	6	1	7
Drug B	2	3	5
Total	8	4	12

An **86% vs 40%** split *looks* dramatic. But the **expected** counts ($\frac{\text{row} \times \text{col}}{12}$) run **1.7 to 4.7**, all **below 5**, so chi-square's approximation no longer holds.

⚠ WATCH OUT

Naïve χ^2 here would report an over-optimistic $p \approx 0.10$, but it's **invalid**: the expected-count ≥ 5 rule is broken.

What Fisher's does instead

Holding all four margins fixed, it reads the **exact** probability of each table straight from the **hypergeometric distribution**: here, the chance Drug A (7 patients) catches **6 of 8** improvers and **1 of 4** non-improvers:

$$P(\text{table}) = \frac{\binom{8}{6} \binom{4}{1}}{\binom{12}{7}} = \frac{112}{792} \approx 0.141$$

Summing this with every **equally- or more-extreme** table gives $p_{2\text{-sided}} \approx \mathbf{0.22} \rightarrow \mathbf{\text{not significant}}$: with only 12 patients, even an 86/40 gap **isn't enough evidence**.

NOTE

Classic for 2×2 ; generalizes to $r \times c$ via **Fisher–Freeman–Halton**. Slightly **conservative**, the **safe default** whenever cells are sparse.

One variable vs a benchmark

The **one-variable** case: hold **one group's** observed frequencies against a *fixed* benchmark, not against another variable. Take **Drug A alone**; how you test depends on **how many categories** the outcome has:

Two categories

One-proportion z-test or **exact binomial**: is the success rate different from a benchmark p_0 ?

*Drug A's **68%** improved vs a known **50%***

Three or more categories

Chi-square goodness-of-fit: do observed counts match an expected pattern?

$$\chi^2 = \sum \frac{(O - E)^2}{E}, \quad \text{df} = k - 1$$

*Drug A graded **improved / no change / worsened** vs a **50 / 30 / 20** profile*



WATCH OUT

Use the **exact binomial** when n is small; the **expected counts ≥ 5** rule applies to goodness-of-fit too. (*Both cases worked out next.*)

One variable vs a benchmark — two categories

Drop Drug B. Take **Drug A alone** and ask a *one-variable* question: is its success rate different from a **fixed external benchmark**, not from another group?

The setup — one drug, one benchmark

Drug A improved **34 of 50 = 68%**. Standard care has a long-run **50%** response. Does A differ from that fixed number?

$$H_0 : p = p_0 = 0.50$$

	Response rate
Drug A (34 / 50)	68%
Standard-of-care benchmark	50%

One-proportion z-test

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.68 - 0.50}{\sqrt{0.25/50}} \approx 2.55$$

Two-sided $p \approx 0.011 \rightarrow$ **reject**: Drug A beats the 50% benchmark. The **95% CI** [0.55, 0.81] excludes 0.50, same verdict.



WATCH OUT

With small n , swap the approximation for the **exact binomial** (here $p \approx 0.015$, same story).



NOTE

Only **one** variable: Drug A's outcome against a *fixed value*. No second variable, so it's **not** a 2×2 independence test.

One variable vs a benchmark — three or more categories

Now grade Drug A's outcome on **three** levels. With $k \geq 3$ categories the proportion test gives way to **chi-square goodness-of-fit**, an observed *profile* against an expected pattern.

Drug A's 50 patients, three ways

Split last slide's 16 non-improvers into *no change* and *worsened*. Does Drug A match the historical **50 / 30 / 20** profile?

Response	Observed	Benchmark	Expected
Improved	34	50%	25
No change	11	30%	15
Worsened	5	20%	10

Chi-square goodness-of-fit

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{9^2}{25} + \frac{4^2}{15} + \frac{5^2}{10} \approx 6.81$$

With $df = k - 1 = 2$, that clears the **5.99** cutoff ($p \approx 0.033$) → **reject**: Drug A's profile **departs** from the benchmark, with more improvement and less worsening.

NOTE

Still **one** variable vs a fixed pattern, not two crossed variables. The expected-count ≥ 5 rule applies; here they're 10–25. ✓

From single tests to regression

A chi-square or Fisher test ends at a single verdict, and the ratios you met earlier (OR, RR) are **fixed summaries** of the table. **Regression goes further: it *models* the outcome against one or more predictors**, so the same Drug A vs Drug B trial yields *adjusted* effect sizes and even patient-level predictions. Which model you reach for depends on the **outcome type**:



Binary outcome → logistic

Did it improve? Yes / no (0/1). Model the **log-odds**
→ read an **odds ratio**.

Coming up: the model · the OR · adjust · dose · predict.



Count outcome → Poisson / NB

How many flare-ups? 0, 1, 2, . . . over time. Model
the **log-rate** → read a **rate ratio**.

Coming up: the model · the RR · exposure · overdispersion.



BOTH FOLLOW THE SAME SHAPE

Each starts from **one row per patient**; with a *single* predictor it merely **reproduces a ratio you already know** (the OR / RR). The real power appears once you add predictors: **adjust** for confounders, take **continuous** inputs, **predict** individuals, or handle **exposure** and **overdispersion**. **Next: the shared dataset both models read.**

The ideal data — one row per patient

Regression doesn't read a 2×2; it reads **one row per patient**. Here's the same trial recorded the ideal way: each patient's drug, covariates, and **two outcomes** (a yes/no *and* a count).

patient	drug	age	severity	dose (mg)	improved	flare_ups	follow_up_mo
1	A	54	mild	40	1	0	6
2	A	61	severe	30	0	2	6
3	B	58	severe	20	0	3	12
...	...	...	...	...	...	...	...
100	B	63	mild	20	0	1	9

Binary column → logistic

`improved` (0/1) is the **logistic** outcome. Grouped by drug it gives back the 2×2 (A 34/16, B 22/28), but the rows also carry **age**, **severity**, **dose**.

Count column → Poisson

`flare_ups` over `follow_up_mo` is the **Poisson** outcome. Note **follow-up varies per patient** (6, 12, 9 ...), and that detail matters later.



NOTE

The aggregated tables you've seen are just **summaries** of this. Regression's power comes from keeping the **individual rows**, which is what lets it adjust for covariates, use continuous predictors, and predict one patient at a time.

Logistic ① — from yes/no to log-odds

The **logistic** model predicts the `improved` column. We want the drug's effect as an **odds ratio (OR)**: how many times higher the *odds* of improving are on A vs B.

Why not predict the probability directly?

Let $p = P(\text{improve})$. A plain line $\beta_0 + \beta_1 x$ can fall **below 0 or above 1**, impossible for a probability. So we model the **log-odds**, free to run from $-\infty$ to $+\infty$:

$$\underbrace{\ln \frac{p}{1-p}}_{\text{log-odds of improving}} = \beta_0 + \beta_1 \text{DrugA}$$

The **logit link** is what guarantees the predicted p stays inside 0–1.

The drug is a 0/1 switch

$\text{DrugA} = 1$ for Drug A, 0 for Drug B:

- **Drug B (off)**: log-odds = $\beta_0 \rightarrow$ the *baseline*
- **Drug A (on)**: log-odds = $\beta_0 + \beta_1$

So β_1 is the **extra log-odds** A adds over B, and e^{β_1} turns it into an **odds ratio**.

NOTE

One predictor here, but the right side can hold **many**: $\beta_0 + \beta_1 \text{DrugA} + \beta_2 \text{age} + \dots$, which is where logistic outgrows a 2×2 (slides ③–⑤).

Logistic ② — fit on the trial → the odds ratio

With the drug as the **only** predictor, the fit just **recovers** each drug's odds from the `improved` column (A 34/16, B 22/28).

The fitted coefficients

Coefficient	Equals	Value
β_0	$\ln(\text{odds}_B) = \ln \frac{22}{28}$	-0.24
β_1	$\ln(\text{OR})$	+0.99

β_0 is Drug B's baseline log-odds; β_1 the extra from switching to A.

Exponentiate to read the odds ratio

$$e^{\beta_1} = \text{OR} = \frac{34/16}{22/28} = 2.70$$

A's odds of improving are **2.7× B's**. 95% CI [1.20, 6.11] excludes 1 ($p \approx 0.017$) → **significant**.



SO WHY NOT JUST USE THE 2×2?

With **one binary predictor**, logistic exactly reproduces the bare odds ratio and adds nothing yet. The payoff appears only when you use the *other columns*: **adjust** for a confounder, take a **continuous** predictor, or **predict** an individual. The next three slides. →

Logistic ③ — real power: adjust for a confounder

The crude OR of **2.70** trusts that the two groups were comparable. They weren't: **Drug B was given to sicker patients**, and severe patients improve less, so part of A's "edge" is really that head start.

`severity` confounds the comparison

of 50	mild	severe
Drug A	30	20
Drug B	15	35

Drug B is **far more severe-heavy** (35 vs 20). Split the $34/16 \cdot 22/28$ table by severity:

stratum	A imp/unch	B imp/unch	OR
mild	27 / 3	13 / 2	1.38
severe	7 / 13	9 / 26	1.56

Adjusted effect ≈ 1.5 , not 2.7

Within **each** severity level the OR is only ~ 1.4 – 1.6 . Pool them (Mantel–Haenszel) for the **severity-adjusted OR**:

$$OR_{\text{adj}} = \frac{\sum a_i d_i / n_i}{\sum b_i c_i / n_i} \approx 1.5$$

Nearly **half** the crude 2.70 was severity, not drug.

NOTE

Stratifying by hand works for one categorical confounder. **Logistic does this automatically** (and for **many** covariates at once) by adding terms: $\beta_0 + \beta_1 \text{DrugA} + \beta_2 \text{severe} + \dots$ Each β is an effect *holding the others fixed*.

Logistic ④ — real power: a continuous predictor

A 2×2 can only hold **categories**. To ask whether **dose** drives improvement you'd have to bin it and throw away detail. Logistic takes `dose_mg` as a **number**; here are the trial's 100 patients grouped by the dose they received:

Improvement climbs with dose

dose (mg)	patients	improved	rate
10	24	6	25%
20	26	12	46%
30	26	18	69%
40	24	20	83%

Model the log-odds as a **line in dose**:

$$\ln \frac{p}{1-p} = \beta_0 + \beta_{\text{dose}} \cdot \text{dose}$$

One slope, read per unit

The fit gives $\beta_{\text{dose}} = +0.092$ per mg. Exponentiate over a **+10 mg** step:

$$e^{10 \beta_{\text{dose}}} = e^{0.92} = \mathbf{2.5}$$

Every **+10 mg multiplies the odds of improving by 2.5×**. Feed a dose back through the model and it returns a probability, e.g. **84%** at 40 mg.

NOTE

A bare odds ratio needs two groups to compare. A **continuous** predictor has none, so logistic summarizes the whole dose–response in a **single slope**.



BRIDGE — COMBINE PREDICTORS IN ONE MODEL

`dose` need not stand alone; it can join the others in **one** fit, $\ln \frac{p}{1-p} = \beta_0 + \beta_1 \text{DrugA} + \beta_2 \text{severe} + \beta_3 \text{age} + \beta_4 \text{dose}$, where each e^{β} is an **adjusted OR** (this is exactly the model on slide ③). Three things to watch:

- **Multicollinearity**: correlated predictors (severe patients are dosed higher) split the shared effect and widen both CIs.
- **Linearity in the logit**: a continuous term is assumed *straight* on the log-odds scale; if it curves, transform or spline it.
- **Interactions**: add a $\text{DrugA} \times \text{severe}$ term to test whether the drug acts *differently* by severity.

Logistic ⑤ — real power: predict one patient

Group odds ratios describe *populations*. The fitted equation also runs **forward**: feed in one patient's covariates → get **their** probability of improving. That's the risk-calculator payoff.

The fitted multivariable model

Putting drug, severity and age in together:

$$\ln \frac{p}{1-p} = 0 + 0.41 \text{ DrugA} - 1.20 \text{ severe} - 0.30 \text{ age}_{10}$$

- 0.41 = $\ln(1.5)$, the **adjusted** drug effect from slide ③
- severe = 1 if severe · age₁₀ = decades above 50

Convert back with $p = \frac{1}{1 + e^{-\eta}}$.

One patient, both drugs

A **60-year-old, severe** patient (age₁₀ = 1, severe = 1):

on	linear predictor η	p
Drug B	$0 - 1.20 - 0.30 = -1.50$	0.18
Drug A	$0.41 - 1.20 - 0.30 = -1.09$	0.25

Switching this specific patient to A lifts their predicted improvement from **18% → 25%**.



NOTE

Same machinery, run in reverse: coefficients **out** for inference, probabilities **out** for prediction. A contingency table gives neither an adjusted effect nor an individual forecast.

Poisson ① — same patients, a count outcome

Now use the `flare_ups` column. A flare can happen 0, 1, 2, ... times, and each patient was followed for a different length of time. A yes/no proportion captures neither, so we model a rate.

Rate = events ÷ person-time

Summed over each drug's varying follow-up (≈ 300 patient-months each):

	flare-ups	patient-months	rate
Drug A	30	300	0.10 / pm
Drug B	60	300	0.20 / pm

The model — a log-rate

$$\ln \mu = \beta_0 + \beta_1 \text{DrugA} + \underbrace{\ln T}_{\text{offset}}$$

- μ = a patient's **expected flare count**
- T = that patient's **follow-up time**
- e^{β_1} = the **rate ratio** (A vs B)

The **log link** keeps the predicted rate positive; the **offset** $\ln T$ turns a raw count into a rate (slide ③).

NOTE

Logistic asked *did you improve?* (one yes/no per patient). **Poisson** asks *how often did it recur?* (a count over time). Same trial, different outcome type.

Poisson ② — the rate ratio, step by step

With the drug as the only predictor, e^{β_1} is just the **ratio of the two rates**.

From rates to a rate ratio

- Drug A: $30/300 = 0.10$ flares per patient-month
- Drug B: $60/300 = 0.20$ flares per patient-month

$$RR = e^{\beta_1} = \frac{0.10}{0.20} = 0.50$$

Drug A **halves** the flare-up rate.

Is it significant?

The standard error of $\ln RR$ comes from the **counts**:

$$SE = \sqrt{\frac{1}{30} + \frac{1}{60}} = 0.224$$

95% CI = $e^{\ln 0.5 \pm 1.96(0.224)} = [0.32, 0.78]$, which excludes 1 ($p \approx 0.002$) → **significant**.



NOTE

Like logistic, the simple one-predictor fit reproduces a number you could get by hand. The real power is the **offset** and the **dispersion** check, in the next two slides.

Poisson ③ — real power: the offset for unequal follow-up

A raw count comparison ("A had 30 flares, B had 60") is **only fair if everyone was watched equally long**. They weren't: `follow_up_mo` varies patient to patient. The **offset** fixes that.

Same count, very different rate

patient	flares	follow-up	rate
X	2	3 mo	0.67 / mo
Y	2	12 mo	0.17 / mo

Identical counts, but X flares **4× faster**. Counting alone would call them equal.

ln T as a fixed offset

Rearranging the model shows the offset converting count → rate:

$$\ln \frac{\mu}{T} = \beta_0 + \beta_1 \text{DrugA}$$

The $\ln T$ term has **no coefficient to fit** (its slope is fixed at 1); it simply credits each patient for their exposure, so the β 's describe **events per unit time**.

NOTE

This is what a bare "30 vs 60" can't do. The offset lets every patient contribute their **own** follow-up, and lets you add covariates on top.

Poisson ④ — real power: overdispersion → negative-binomial

Poisson makes one strong promise: **variance = mean**. Real flare counts break it: a few patients flare constantly, most never do. Ignore that and the CIs come out **too narrow**.

Check the dispersion

Drug A's 50 patients (30 flares total) are **clumpy**:

	value
mean count	0.60
variance	1.27
dispersion = var ÷ mean	≈ 2.1

Poisson assumes this ratio is **1**. At 2.1 the data are **over-dispersed**.

Negative-binomial — honest CIs

NB adds a second parameter for the extra spread. It keeps the **same rate ratio (0.50)** but **widens the CI**, so you don't over-claim significance.

- variance > mean → **negative-binomial**
- excess zeros (many never-flare) → **zero-inflated**



WHY IT MATTERS

Under-estimating the spread makes a Poisson fit report **false confidence**, a too-small p . The dispersion check is the count-model version of *testing your assumptions before trusting the result*.

Wrap-up: One trial, four questions — pick the tool

The whole section ran on **one Drug A vs Drug B trial**, and the *question* picks the tool.

Single tests (left) answer *is there an effect?* with a *p*-value.

Models (right) go further, to an **effect size** you can *adjust* and *predict* from.

Across all four, **ratio = 1** means **no difference**.

① Two variables linked?

Chi-square independence

The $r \times c$ table $\rightarrow \chi^2, p$,
then **Cramér's V** & residuals
(*how strong, which cells*).

Fisher's exact

Tiny / sparse cells: an exact
p.

② One sample vs a benchmark?

One-proportion / exact binomial

2 categories vs a fixed
 p_0 .

Chi-square goodness-of-fit

3+ categories vs an
expected pattern.

③ Model the odds? (binary)

Logistic regression

Improved yes / no $\rightarrow$ an
odds ratio.

Add covariates to
adjust, take a
continuous input,
predict one patient.

④ Model the rate? (counts)

Poisson / negative-binomial

Flare-ups per exposure
 $\rightarrow$ a **rate ratio**.

An **offset** handles
unequal follow-up; **NB**
when over-dispersed.

Self-check

A trial records **improved / not improved** for two drugs. One drug has a cell with only **3 patients**, and a reviewer asks you to **adjust the comparison for patient age**. Which test gives the raw association, and which one lets you adjust?



ANSWER

With a sparse cell (expected count < 5), use **Fisher's exact test** for the raw association rather than chi-square. To **adjust for age**, fit a **logistic regression** (improved yes/no $\sim$ drug + age); the drug coefficient is an **age-adjusted odds ratio**.

■ PART 6 · RANK-BASED CATEGORIES

Ranking categories

When categories carry an **order** but no fixed spacing, you don't count them. You **rank** them.

Ordinal data: order without spacing

A 5-point Likert answer (*Strongly disagree* → *Strongly agree*) has a clear **order**, but the **gaps are unknown**: the step from *Disagree* to *Neutral* isn't the same "distance" as *Agree* to *Strongly agree*.



No number to average

"The **mean satisfaction is 3.4**" treats the codes 1–5 as real numbers, but the spacing isn't real, so the average **isn't either**.



So rank instead

The **order** is real. Sort the responses and work with each one's **rank**, since the median and rank-based tests keep only what the data actually supports.

1 *Strongly disagree* · 2 *Disagree* · 3 *Neutral* · 4 *Agree* · 5 *Strongly agree*



NOTE

This is the **rank-based branch** for **ordinal** data that Part 5 flagged (and Part 2 introduced): a mean is meaningless, but the **order** lets a whole family of tests work on the **ranks**, not the raw codes.

The shared engine: turn values into ranks

All three rank tests start the same way: **pool every observation, sort, and assign ranks** $1 \dots N$. Tied values share the **average** of the ranks they'd occupy.

Six responses → six ranks

Responses **4, 2, 5, 4, 3, 2**, sorted and ranked:

sorted	2	2	3	4	4	5
position	1	2	3	4	5	6
rank	1.5	1.5	3	4.5	4.5	6

The two **2**'s would take positions 1–2 → both get $\frac{1+2}{2} = 1.5$;
the two **4**'s take 4–5 → both get 4.5.

Why ranks?

- **Distribution-free:** no bell curve assumed, so skew and outliers don't break it.
- **Works on ordinal:** needs only an *order*, not real distances.
- **Robust:** one extreme answer moves a rank by one step, not by its raw magnitude.

NOTE

The same trick lets these tests run on **non-normal continuous** data too (Part 5's table listed them under *both* ordinal and interval). You just rank the numbers and drop the interval detail.

The ordinal toolkit: two columns, by question

Pick the **question** on the left, then read across to the **rank test** (a quick, distribution-free p) and its **model** counterpart (an adjustable, predictive effect). The same jump from single tests to models that Part 5 made.

Your question	Non-parametric — rank-based	Parametric — model-based
Compare 2 groups	Mann–Whitney U	ordinal regression (<i>binary predictor</i>)
Compare 3+ groups	Kruskal–Wallis → Dunn	ordinal regression (<i>group factor</i>)
Measure a relationship	Spearman's ρ (Kendall's τ)	ordinal regression (<i>continuous predictor</i>)



TIP

- **Rank tests** answer *is there a shift / a trend?* and give a p -value plus an effect size.
- **Ordinal models** (*proportional-odds logistic · probit*) go further: one **odds ratio** or latent shift you can **adjust** for covariates and **predict** from.

The whole part builds left → right: a quick test first, a richer model when you need more.

The running survey — one row per respondent

A course is rated "*I'd recommend it*" on the 5-point Likert by 120 students, 40 in each format (*Online · Hybrid · In-person*), with each student's **prior experience** also recorded.

One row per respondent

student	format	experience	sat.
1	In-person	proficient	5
2	Online	novice	2
3	Hybrid	some	4
...	...	...	...
120	Online	novice	1

...the satisfaction × format counts

level	Onl.	Hyb.	In-p.	total
1 Strongly disagree	8	4	2	14
2 Disagree	12	8	4	24
3 Neutral	10	10	8	28
4 Agree	7	12	14	33
5 Strongly agree	3	6	12	21



NOTE

The counts only **summarize** the rows. Like Part 5's regression, the **ordinal models** later read the *full rows*, which is what lets them adjust for **experience** and predict one student.

Mann–Whitney U ① — the idea: a tournament of duels

You can't average "Agree" and "Disagree." So Mann–Whitney never adds the scores. Instead it stages a **tournament**: every member of one group **duels** every member of the other, and you simply **count the wins**.

Every cross-group pair duels once

A tiny **2-vs-2** warm-up: **Group A** rated **2**, **4** · **Group B** rated **3**, **5**. Pair **every A with every B** → $2 \times 2 = 4$ duels; the **higher rating wins** (a tie pays each side ½):

duel	winner
A 2 vs B 3	B
A 2 vs B 5	B
A 4 vs B 3	A
A 4 vs B 5	B

U = a group's win count

Group B won **3** of the 4 duels, Group A won **1**:

- $U_B = 3$ · $U_A = 1$
- A bigger U just means *that group rates higher more often*.
- The two always add to the number of duels: $3 + 1 = 4$.



UNEQUAL GROUPS? STILL FAIR

Duels are always **A vs B**, never a group against itself, so there are $n_A \times n_B$ of them, e.g. **3 vs 2** → **6 duels**. A bigger group gets **no edge**, because U is read as a **rate** (wins ÷ duels), not a raw total.



NOTE

That's the whole engine: **count cross-group wins**. Now run it on the real survey (**6 Online vs 6 In-person** → $6 \times 6 = 36$ duels) with a shortcut so we never list all 36 by hand.

Mann–Whitney U ② — rank: the shortcut to counting wins

Listing all 36 duels by hand is tedious. **Ranking does it for free:** pool both groups, sort low → high, and each value's rank already says how high it sits. Then add up each group's ranks.

Pool all 12, sort, rank

Online: 1, 2, 2, 3, 3, 4 · **In-person:** 2, 3, 4, 4, 5, 5

Number the 12 sorted positions; tied values share the average position:

value	1	2	3	4	5
sorted positions	1	2–4	5–7	8–10	11–12
rank (avg)	1	3	6	9	11.5

Sum the ranks in each group

- **Online** ranks 1, 3, 3, 6, 6, 9 $\Rightarrow R_{\text{Online}} = 28$
- **In-person** ranks 3, 6, 9, 9, 11.5, 11.5 $\Rightarrow R_{\text{In-person}} = 50$

Check: all 12 ranks total $\frac{12 \cdot 13}{2} = 78$, and $28 + 50 = 78 \checkmark$



WHY RANKING = COUNTING WINS

A student's **rank** measures how many students they out-rate. So In-person's bigger rank-sum (**50 vs 28**) means its students sit **higher in the pool**, so they **win more duels**. Next we read off the exact win-count.

Mann-Whitney U ③ — from ranks to the U statistic

A group's rank-sum also counts it beating **its own** members. Subtract that fixed part, $\frac{n(n+1)}{2}$, and what's left is exactly the **cross-group wins**, the **U** from slide ①.

Strip out the within-group part

$$U = R - \frac{n(n+1)}{2}$$

Each group has $n = 6$, so $\frac{6 \cdot 7}{2} = 21$:

- $U_{\text{In-person}} = 50 - 21 = \mathbf{29}$
- $U_{\text{Online}} = 28 - 21 = 7$

Check: $29 + 7 = 36 = 6 \times 6 \checkmark$, the very duel-count from slide ①.

The two are redundant; the test reports the **smaller**, $U = 7$.

One number, two questions

We now have $U_{\text{In-person}} = 29$ of 36, and it answers **two separate questions** beginners often blur:



HOW BIG IS THE LEAD?

Effect size: does In-person rate higher *by a lot*? → slide ④



HOW SURE CAN WE BE?

Significance: could **luck** alone fake this gap? → slide ⑤



ONE TEST, TWO NAMES

U is just a fixed shift of the rank-sum R , so they hold the **same information**. Wilcoxon (1945) summed ranks; Mann-Whitney (1947) counted wins, but it's **one test**.

Mann-Whitney U ④ — how big is the lead? (effect size)

Question 1: how big? Turn the win-count into one plain number, the **probability of superiority**: pick one student from each group at random, how often does In-person win?

| **Win-rate = effect size**

$$A = \frac{U_{\text{In-person}}}{n_1 n_2} = \frac{29}{36} \approx \mathbf{0.81}$$

Use In-person's **own** win-count $U = 29$ (not the smaller 7 we report for the test). So a random **In-person** student out-rates a random **Online** one about **81%** of the time.



Read it like a coin

0.50 = a coin-flip, no difference. **0.81** is a large, clear lead, stated in plain words, no jargon.



SIZE ≠ CERTAINTY

Effect size says **how big** the gap is, not **how sure** we are. A true 81% stays 81% with 6 students or 600. Whether **6 is enough to trust it** is the next, separate question.

Mann-Whitney U ☹ — could it be luck? (the chance test)

Question 2: how sure? An 81% lead is big, but with only 6 + 6 students, could plain **chance** produce it? The test: compare our result against **every split that chance could make**, step by step.

Build every split chance could make

1 • Suppose the two groups **really scored the same**. Then the labels *Online / In-person* are just **stickers** dropped at random onto the 12 ranked students.

2 • Count every way to pick **which 6 students are labelled "In-person"** (the **other 6 are Online automatically**):

$$\binom{12}{6} = \frac{12!}{6!6!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 924$$

→ **924 equally-likely splits**, the full list of what **pure chance** could deal. We need this total to judge **how rare our 29 really is**.

3 • In every split, count how many of the **36 duels In-person wins**.

Most splits land near a tie (18)

	In-person wins	how many splits (924)	
1	18 tie	58	most common
2	24	39	
3	28	18	
4	29 ← ours	13	rare
5	31	7	
6	36 all-win	1	rarest

Each bar **counts how many of the 924 splits** give that win-count. If matched, In-person lands **near 18**; a real lead pushes it **higher**, and our **29** sits far out in the **rare zone**. Rare *enough* to rule out chance? Slide ☹.

These 924 tallies depend only on "**6 vs 6**", not on the ratings, so one table fits every 6-vs-6 study.

Mann–Whitney U ⑥ — read the verdict (the 5% line)

The **p-value** asks: *of all 924 splits, what share are at least as extreme as ours?* Compare that to the **5% line** researchers use to call a result "real."

How rare is our result?

Sum every split from **29 up** (our result, or more extreme):

In-person wins	29	30	31	32	33	34	35	36	≥ 29
splits (of 924)	13	11	7	5	3	2	1	1	43

- **43 of 924** splits are at least as extreme → $43/924 \approx 0.047 \approx 5\%$ on **one side**.
- A real gap could favour **either** group, so we also count the mirror end → $p \approx 0.09$ (a **two-sided** test).

NOTE

"At least as extreme" includes outcomes **even more one-sided** than ours: if 29 counts as evidence, 30 or 31 counts too.

Where does the 5% line fall?

"Real" (**significant**) means $p \leq 5\%$. Climb the win-counts until p first drops under 5%:

In-person wins ≥	p (both ends)	under 5%?
29 — ours	9.3%	×
30	6.5%	×
31	4.1%	✓

So the 5% line sits at **31 wins**, the same as "smaller $U \leq 5$." To pass, In-person must win **almost all 36** duels.



BIG EFFECT, TOO FEW PEOPLE — "UNDERPOWERED"

Our **29** is just **2 wins short** of the 31 needed, so $p \approx 0.09 > 0.05$: a real-looking **81% lead, but not proven yet**. Flip two duels and it would pass: **6 per group is simply too few**. The full **40-per-group** survey has the **power** to confirm the shift. Next, with **Kruskal–Wallis**.

Kruskal–Wallis ① — the idea: each group vs. the middle

Mann–Whitney pitted **two** groups in duels and gave one win-count U . With **three** formats there's no single "win count", so Kruskal–Wallis asks a sister question: **how far does each group's *average rank* sit from the middle?**

A tiny warm-up: 12 students, 3 groups

Rank all 12 low $\rightarrow$ high, **ignore the labels**, then average each group's ranks:

group	ranks	mean rank
Group A	1, 2, 3, 4	2.5
Group B	5, 6, 7, 8	6.5
Group C	9, 10, 11, 12	10.5

The **middle rank** is $\frac{N+1}{2} = \frac{13}{2} = \mathbf{6.5}$, where *every* group lands if the grouping makes no difference.

Spread from the middle = the signal

- If the groups were really the same, high and low ranks would **mix evenly**, so each mean rank sits near **6.5**.
- A real difference makes one group **hog the top ranks** (mean climbs **up**) and another the bottom (mean drops **down**).

Group A (**2.5**) and Group C (**10.5**) sit a full **4 ranks** either side of 6.5. That is a wide spread, so something is going on.



ONE NUMBER FOR THE WHOLE SPREAD

Kruskal–Wallis rolls "how far is each group from the middle?" into a **single statistic H** , big when the mean ranks fan out and near zero when they cluster. Next: the real survey, then H itself.

Kruskal-Wallis ② — rank all 120, average each group

Now the real survey: $N = 120$, 40 per format. Pool everyone, sort low $\rightarrow$ high, give tied levels their block's **mid-rank**, then average the ranks within each format.

Each Likert level fills a block of ranks

Sorted across all 120, each rating occupies a run of positions; its **mid-rank** is that run's midpoint:

level	1	2	3	4	5
positions	1–14	15–38	39–66	67–99	100–120
mid-rank	7.5	26.5	52.5	83	110

So every student who answered **3** (positions 39–66) gets rank **52.5**.

Mean rank per format vs. the middle

Each format's **mean rank** = its rank-sum $\div$ 40:

format	mean rank
Online	45.4
Hybrid	60.6
In-person	75.6

The overall middle is $\frac{N+1}{2} = \frac{121}{2} = \mathbf{60.5}$. The means **climb monotonically** Online $\rightarrow$ In-person. That is exactly the spread slide ① called the signal, now on real data.



NOTE

Online sits **~15 below** the middle, In-person **~15 above**. Slide ③ turns that spread into H .

Kruskal–Wallis ③ — calculate H: the spread in one number

H adds up how far each mean rank strays from the middle (squared, and weighted by group size), then rescales it. Plug in slide ②'s three mean ranks (middle $\bar{R} = 60.5$):

$$\begin{aligned} H &= \frac{12}{N(N+1)} \sum_i n_i (\bar{R}_i - \bar{R})^2 \\ &= \frac{12}{120 \cdot 121} \left[40(45.4 - 60.5)^2 + 40(60.6 - 60.5)^2 + 40(75.6 - 60.5)^2 \right] \approx \mathbf{15.1} \end{aligned}$$

Read it piece by piece

- $(\bar{R}_i - \bar{R})^2$: each group's **distance from the middle**, *squared* so above and below both count as "off".
- $n_i \times$: **weight by group size**, since a bigger group straying counts for more.
- $\frac{12}{N(N+1)}$: a **scaling constant** so that, with no real difference, H lands at a *known* size (a χ^2), letting us read off rarity.

The textbook **computing form** is the same thing rearranged:

$$H = \frac{12}{N(N+1)} \sum_i n_i \bar{R}_i^2 - 3(N+1)$$

Ties, and the tie to Mann–Whitney

With only **5 distinct levels** the data is heavily **tied**, which shrinks the raw H a little. Slide ④ works the tie-correction through to the final value.



SO WHY NO H IN MANN–WHITNEY?

H is just the **general** version of "are the ranks too lopsided?" Run Kruskal–Wallis on exactly **two** groups and it is Mann–Whitney. In fact, $H = z^2$. Two groups need no H because the single win-count U already tells the whole story.

Kruskal-Wallis ④ — every number, worked end to end

Every figure traces back to one table, the **survey counts**: a format's **mean rank** weights the level mid-ranks (7.5, 26.5, 52.5, 83, 110) by its counts, and the **tied** column feeds the tie-correction.

The data: survey counts

level	Onl.	Hyb.	In-p.	tied t_j
1 Strongly disagree	8	4	2	14
2 Disagree	12	8	4	24
3 Neutral	10	10	8	28
4 Agree	7	12	14	33
5 Strongly agree	3	6	12	21

Each **tied** t_j is the row summed across formats (8 + 4 + 2 = 14, and so on), so $N = 120$ total.

Counts → raw H → corrected H

Weight the mid-ranks by **Online's** counts:

$$\bar{R}_{\text{Onl}} = \frac{8(7.5) + 12(26.5) + 10(52.5) + 7(83) + 3(110)}{40} = 45.4$$

Hybrid **60.6**, In-person **75.6** likewise; slide ③'s spread formula gives the **raw** $H \approx 15.1$. Then correct for the ties:

$$C = 1 - \frac{\sum_j (t_j^3 - t_j)}{N^3 - N} = 1 - \frac{83,598}{1,727,880} \approx 0.95$$

$$\text{So } H = \frac{15.1}{0.95} \approx \mathbf{15.9}.$$

NOTE

Dividing by $C < 1$ **lifts** H (ties cost rank-variance, the correction restores it). Slide ⑤ tests this $H \approx 15.9$ against the 5% line.

Kruskal-Wallis ⑤ — verdict, then which pairs

From slide ④ we have $H \approx 15.9$. Compare it to a **chi-square** with $df = k - 1$ degrees of freedom ($k = 3$ groups $\rightarrow df = 2$).

The omnibus result

$H \approx 15.9$ with $df = 2$ clears the **5.99** cutoff (the χ^2_2 5% value) easily $\rightarrow p < 0.001$. **Reject**: at least one format's ratings differ.

Effect size rescales H onto a fixed **0-to-1 scale**:

$$\varepsilon^2 = \frac{H}{N - 1} = \frac{15.9}{119} \approx 0.13$$

Reading ε^2 : 0 means identical groups, 1 means perfectly separated. So 0.13 is about **13% of the rank variation explained by format** (the rank cousin of R^2), a **small-to-moderate** effect that compares across any sample size.

Omnibus only, then post-hoc

Like ANOVA, H says *some* group differs, **not which**. Follow up with **Dunn's test** on the pairwise rank differences, with a multiplicity correction (Bonferroni / Holm), the rank-based analog of Tukey.



WATCH OUT

Don't run three separate Mann-Whitneys instead, because that re-inflates the false-positive rate Part 4 warned about. The omnibus *then* Dunn keeps it controlled.

Spearman's ρ ① — a monotonic relationship

The third question: do two ordered variables move together? Take prior experience (years) vs a satisfaction score for 8 students. Spearman is just Pearson's correlation computed on the ranks.

Rank each variable, then compare

Rank each column **low** $\rightarrow$ **high** (r_x, r_y); the gap is $d = r_x - r_y$:

student	exp.	sat.	r_x	r_y	d	d^2
A	3	40	1	1	0	0
B	5	55	2	3	-1	1
C	6	50	3	2	1	1
D	8	65	4	4	0	0
E	10	70	5	6	-1	1
F	12	68	6	5	1	1
G	14	85	7	7	0	0
H	16	90	8	8	0	0

The rank-difference formula

With **no ties**, ρ reads straight off the squared rank gaps d^2 :

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6 \cdot 4}{8 \cdot 63} \approx \mathbf{0.95}$$

Only two near-neighbour swaps ($\sum d^2 = 4$) $\rightarrow$ a **strong positive** monotonic link: more experience tracks higher satisfaction.

Spearman's ρ ② — reading it, and Kendall's τ

ρ runs from -1 (perfectly reversed) through 0 (no monotonic link) to $+1$ (perfectly aligned).



Monotonic, not linear

ρ only asks whether y keeps **rising** (or falling) as x rises, not whether the rise is a straight line. A curved-but-always-increasing relationship can score $\rho = 1$, where Pearson's r would dip.



Significance

For $n = 8$, $\rho \approx 0.95$ clears the 5% critical value (≈ 0.74)
→ the correlation is **significant** ($p < 0.001$). Larger n uses a t or permutation test.



ADVANCED — KENDALL'S τ

A rank correlation built from **concordant minus discordant pairs**: of all student pairs, how many agree on the ordering vs disagree. It's **more robust with many ties** and has a cleaner probabilistic reading, though it usually runs numerically smaller than ρ . With the **5-point Likert** (heavy ties) you'd compute Spearman as **Pearson on the mid-ranks**, or reach for τ .

From rank tests to ordinal models

A Mann–Whitney or Kruskal–Wallis ends at a **p-value plus an effect size**. It can't *adjust* for a confounder, take a *continuous* predictor, or *predict* an individual. For that, **model the ordinal outcome directly**, exactly as Part 5 moved from chi-square to logistic regression.



Logit link → proportional-odds

Model the **cumulative log-odds** of being in a higher category → read a single **odds ratio** that shifts the whole rating distribution.

Coming up: cut-points · the OR · adjust & predict.



Probit link → ordered probit

Same cut-point idea, but assume a hidden **normally-distributed** "true satisfaction" sliced into the 5 bands → read a **latent shift** in SD units.

Coming up: the latent-variable view.



BOTH RESPECT THE ORDER

Neither model treats the 5 levels as unordered labels (that would be **multinomial** regression, throwing the order away) nor as real numbers (that would be **linear** regression, inventing spacing). They model the **thresholds between** ordered categories, the honest middle path.

Proportional-odds ① — cumulative cut-points

Don't model each of the 5 categories separately. Instead, at each of the **4 thresholds**, model the **cumulative** log-odds of falling *at or below* it, and let **one slope** β describe the predictor at **every** threshold.

One slope, four intercepts

$$\ln \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j - \beta x, \quad j = 1 \dots 4$$

- Y is the **rating** (1–5); x the **predictor** (here In-person vs Online).
- α_j is a **separate cut-point** for each threshold (they only set *where* the categories split).
- β is **shared** across all four: the predictor shifts the *whole* distribution by the same amount.

Why cumulative?

Splitting at " $\leq$ *Neutral* vs *better*" uses the **order**. Every threshold respects that $1 < 2 < \dots < 5$. A single β then means "**proportional odds**": the predictor's effect is the **same multiplicative shift** at "*better than Disagree*", "*better than Neutral*", and so on.



NOTE

The cut-points slide the boundaries; β slides the **student**. That separation is what lets one number summarize an ordinal effect.

Proportional-odds ② — one odds ratio

Fit on the survey (In-person vs Online). The **cumulative odds of a higher rating** are larger for In-person at **every** threshold, by roughly the **same factor**.

The cumulative odds of $Y > j$

Odds of $Y > j$ = students *above* the cut ÷ those *at-or-below* (from the counts); **OR** = In-person ÷ Online. (*Online > Disagree*: $20 \div 20 = 1.0$.)

threshold $Y > j$	Online	In-person	OR
> Strongly disagree	4.0	19.0	4.8
> Disagree	1.0	5.7	5.7
> Neutral	0.33	1.9	5.6
> Agree	0.081	0.43	5.3

All four ORs **hover around 5**, the proportional-odds assumption in action.

Collapse them to one number

The model replaces those four with a single fitted effect:

$$e^{\beta} = \text{OR} \approx \mathbf{5.3} \quad (\beta \approx 1.67)$$

In-person students have **~5× the odds** of landing in a *higher* satisfaction category, at every cut. 95% CI $\approx [2.8, 10]$ excludes 1 $\rightarrow p < 0.001$.



NOTE

This is the effect size Mann–Whitney couldn't give: a single interpretable, testable number for an ordinal shift.

Proportional-odds ③ — adjust & go continuous

With a single binary predictor, the OR just **re-expresses the table**. The payoff (exactly as in Part 5) is using the **other columns**: adjust for a confounder, or take a predictor as a **number**.

Adjust for a confounder

More-**experienced** students both **chose In-person** *and* rate higher anyway, so part of the crude **5.3** is really experience.

Add a term: $\alpha_j - \beta_1 \text{InPerson} - \beta_2 \text{experience}$

Each e^β is now an **adjusted** OR, the format effect *holding experience fixed* (here it falls to ≈ 3.5). The model stratifies automatically, over many covariates at once.

Take a continuous predictor

A rank test can't use **study-hours** as a number; the model takes one slope:

$$\ln \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j - \beta_{\text{hrs}} \cdot \text{hours}$$

$e^{\beta_{\text{hrs}}}$ is the OR of a *higher rating* per extra hour.



WATCH OUT

Watch the same three as Part 5's logistic: **collinearity** (experience $\leftrightarrow$ format), **linearity** of a continuous term on the log-odds scale, and **interactions**.

Proportional-odds ④ — predict one student's profile

Run the fitted model **forward**: feed a student's covariates → get their **whole rating distribution**, not just one number. The cut-points turn the linear predictor into probabilities across all five levels.

Cut-points + the In-person shift

Start from the baseline (Online) cut-points with $\beta = 1.67$. Each α_j is just $-\ln$ of Online's odds from ② ($-\ln 4.0 = -1.39$, $-\ln 1.0 = 0$, ...):

α_1	α_2	α_3	α_4
-1.39	0	1.10	2.51

For an **In-person** student, each cumulative logit is $\alpha_j - 1.67$, then

$$P(Y \leq j) = \frac{1}{1 + e^{-(\alpha_j - \beta)}}.$$

Differencing gives each category

$$P(Y = j) = P(Y \leq j) - P(Y \leq j-1):$$

level	1	2	3	4	5
prob	.05	.11	.20	.34	.30

This In-person student is **most likely to Agree** (0.34), with **0.30** on *Strongly agree*, a full predicted profile (sums to 1).

NOTE

Coefficients **out** for inference, a **category distribution out** for prediction. Neither a rank test nor a contingency table offers this.

The proportional-odds assumption — and probit

The model's power rests on **one slope for all thresholds**. Check it; and meet the sibling model that swaps the link.

When parallel lines fail

If a predictor's effect **differs by threshold** (e.g. it lifts *Strongly agree* but not *Disagree*), the single OR is misleading. Detect it with a **Brant** or score test of the parallel-lines assumption.

Fixes when it fails:

- **Partial proportional odds**: free the slope only for the offending predictor.
- **Multinomial** logistic: drop the order entirely (last resort).

Probit — a latent-normal cut

Same cut-point model, but the **normal-CDF** link: imagine a hidden, normally-distributed "*true satisfaction*" sliced into 5 bands.

- $\beta_{\text{probit}} \approx 0.95 \rightarrow$ In-person sits **~1 SD** higher on that latent scale.
- No tidy odds ratio: coefficients are **z-shifts**, in latent SDs.
- $\beta_{\text{logit}} \approx 1.7 \beta_{\text{probit}}$: the fits agree, only the **link** differs.



NOTE

Use **logit** for an odds ratio; **probit** when a latent-normal story (or econometric convention) fits. The verdict is the same.

Wrap-up: one ordinal survey, pick the tool

The whole part ran on **one Likert survey** (*Online · Hybrid · In-person*). **Rank tests** (left) give a fast, distribution-free p + effect size; **ordinal models** (right) give an **adjustable, predictive** effect. The question picks the tool.

① Two groups differ?

Mann–Whitney U

Pooled ranks $\rightarrow U$; the **probability of superiority** U/n_1n_2 as effect size. = Wilcoxon rank-sum.

② 3+ groups differ?

Kruskal–Wallis

Mean ranks $\rightarrow H \sim \chi^2_{k-1}$; ε^2 effect size; then **Dunn's** post-hoc for the pairs.

③ A relationship?

Spearman's ρ (*Kendall's τ*)

Pearson on the ranks; **monotonic**, not linear.

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2-1)}.$$

④ Model the ordinal outcome

Proportional-odds / probit

One **OR** (or latent SD shift) across all cut-points: **adjust**, go **continuous**, **predict** a profile.

Self-check

A reviewer has **patient-satisfaction** scored *poor* / *fair* / *good* / *excellent* across **three clinics**, and asks you to (a) test whether the clinics differ and (b) report the clinic effect **adjusted for patient age**. Which test answers each, and what must you check for (b)?



ANSWER

Satisfaction is **ordinal**, so (a) use **Kruskal–Wallis** across the three clinics. A significant H means at least one differs (follow with **Dunn’s test** for the pairs). For (b), a rank test can’t adjust, so fit a **proportional-odds (ordinal) logistic** model, `satisfaction ~ clinic + age`, where each clinic coefficient is an **age-adjusted odds ratio** of a higher satisfaction category. Before trusting it, **check the proportional-odds (parallel-lines) assumption** (Brant test); if it fails, use **partial proportional odds**. Prefer an **ordered probit** if you’d rather read the effect as a latent-normal SD shift.

• PART 7 · HYPOTHESIS TESTING

Hypotheses, significance & error

The machinery of "is this difference real?" and the two ways it can mislead you.

Null vs. alternative hypothesis

Every test pits two statements against each other. By convention you start from the **null** (the assumption of *no effect*) and ask whether the data give you enough reason to abandon it.



Null hypothesis (H_0)

There is **no difference** / no effect. Treatment 1 = Treatment 2. This is what you try to disprove.



Alternative hypothesis (H_1)

There **is** a difference / an effect. If the data are convincing, you **reject** H_0 and accept H_1 .



WORKED EXAMPLE — MUSCLE TRAINING

A new training method is proposed.

H_0 : the new method has no greater effect than existing ones. H_1 : the new method has a greater effect.

Framing a hypothesis

The hypothesis must be stated **before** you analyze, and phrased so the null can be cleanly negated.



WORKED EXAMPLE — A REMEDY FOR BLOOD SUGAR

General claim: Remedy X affects glucose levels.

- H_0 : Remedy X has **no** effect on glucose levels.
- H_1 : Remedy X **does** affect glucose levels.



TIP

Notice the pattern: the null is always the *boring* version, "nothing happened." Your evidence has to earn the right to reject it.

Three trial designs

What you're trying to *show* changes how you frame the comparison, common in clinical trials.



Superiority

The new treatment is **better** than the standard (or placebo).



Equivalence

The new treatment is **as good as** the standard: no meaningful difference.



Inferiority

The new treatment is **less effective** than the standard.



NOTE

Most studies aim for **superiority**, but "is the cheaper drug *equivalent*?" is an equally valid (and common) question.

The significance level, α

α (alpha) is the threshold you set *in advance* for rejecting the null. The convention is $\alpha = 0.05$.

It is the probability of **falsely rejecting a true null**, calling a difference real when it isn't. That mistake is **Type I error**.

- $\alpha = 0.05 \rightarrow$ accept a 1-in-20 false-positive rate
- $\alpha = 0.01 \rightarrow$ more cautious, 1-in-100
- Lower $\alpha \rightarrow$ fewer false positives

■ $\alpha = 0.05$ means 5 in 100 by chance

1	.	.	.	.	.	.	.	.	.	.
2	.	.	★	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	★	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	★	.	.	.	.	.	.	.	.
7	.	.	.	.	.	.	.	★	.	.
8	.	.	.	.	.	.	.	.	.	.
9	.	.	.	★	.	.	.	.	.	.
10	.	.	.	.	.	.	.	.	.	.

★ = a "significant" result that's really just chance.



WATCH OUT

α must be fixed **before** you look at the results; choosing it afterward is moving the goalposts.

One-tailed vs. two-tailed

Where you place the "rejection region" depends on whether you can predict the *direction* of the effect.

One-tailed

Use when the direction is **known in advance**: you only care if the new drug is *better*.



Two-tailed

Use when the effect could go **either way**: you can't predict the direction.



 Two-tailed is the safer default; reach for one-tailed only with a genuine, pre-stated directional prediction.

The p-value

The **p-value** is the probability of seeing a result *at least this extreme* if the null were true (i.e. by chance alone).

Decision rule: if $p < \alpha$, reject H_0 and accept H_1 , and the result is "statistically significant." If $p \geq \alpha$, you fail to reject H_0 .



Small p

A result this extreme would be rare by chance → evidence against H_0 .



Large p

A result this ordinary happens easily by chance → no grounds to reject H_0 .

A p-value is **not** "the probability the null is true," and it says nothing about how *big* the effect is.

What does a p-value mean?

Behind the formula is one plain-language question:

If there were really **no effect**, how likely would it be to see a result like ours (**or an even stronger one**) purely by chance?



A worked example

- **Group A** takes the medicine
- **Group B** does not
- Group A improves by **10 points more** than B

That **10-point gap** is the *observed result*.



The reasoning, step by step

```
1  Assume the medicine has NO effect   (the null)
2                                     ↓
3  Observed:  A beat B by +10 points
4                                     ↓
5  Ask:   could a gap this big be chance alone?
6                                     ↓
7  p-value:  how likely that would be
```



Even with **no real effect**, the two groups still differ a little, because people naturally vary. The p-value asks whether **+10** is more than that everyday wobble.

Reading the p-value

Same example, two possible verdicts: the p-value tells you which is more plausible.



Small p-value

A **+10 gap** would be **unusual** if the medicine truly did nothing.

- Evidence the medicine **may have a real effect**.
- Small p-value = **stronger evidence for a real effect**.



Large p-value

A **+10 gap** could **easily happen** by chance, even with no effect.

- **Not enough evidence** to claim a real effect.
- Large p-value = **not enough evidence**.



A large p-value is **not** proof the medicine does nothing.
It only means this study failed to show a clear effect. (More on that next.)

Two ways to be wrong

A test makes a **decision**, but reality may disagree. There are exactly two ways that can go wrong.



The setup

H_0 = the medicine does nothing. Your test returns one of two verdicts:

- **Reject H_0** → "the medicine probably works"
- **Fail to reject H_0** → "not enough evidence it works"



Reality can disagree

The drug either truly works or it doesn't — whatever we decide.

When verdict and reality **disagree**, it's a **false positive** (Type I) or a **false negative** (Type II).

Type I error — a false positive

The medicine does **nothing**, but we conclude it **works** — we **reject H_0** even though it's true.



What happens

The drug is useless, but **random luck** let Group A improve more than Group B.

We read that gap as proof and **back a useless drug**.



In one line

Seeing an effect that isn't really there. A false alarm.



ITS PROBABILITY IS A

α is the false-positive risk you set **in advance** — usually **0.05, 0.01, 0.001**.

At $\alpha = 0.05$ you accept a **5% chance** of claiming an effect that isn't there.

Type II error — a false negative

The medicine **works**, but the study **fails to show it** — we fail to reject H_0 even though it's false.



What happens

The drug truly helps, but the result wasn't clear enough, often because the study was too small.

We call it **not enough evidence** and **throw away a helpful drug**.



In one line

Missing an effect that is really there. A missed discovery.



ITS PROBABILITY IS β — AND POWER = $1 - \beta$

β is the chance of missing a real effect. **Power** = $1 - \beta$ is the chance of catching it. High power rarely misses a real effect, but low power often does. (More on power soon.)

The whole picture, in plain words

Reality × your decision gives four outcomes — two right, two wrong.

H_0 true — drug does nothing

H_0 false — drug works

Reject H_0 (say it works)

✗ Type I — false positive (α)

✓ Correct — power ($1 - \beta$)

Fail to reject H_0 (say not enough evidence)

✓ Correct

✗ Type II — false negative (β)



TYPE I — FALSE ALARM

Believe something's real when it isn't.

You back a useless drug.



TYPE II — MISSED DISCOVERY

Something's real, but you miss it.

You throw away a helpful drug.



Lowering one usually raises the other: tighten α (fewer false positives) and β climbs (more misses). Good design balances the two.

What is the opposite of "a significant difference"?

It is **not** "there is no difference."

It is: "**we cannot say there is a difference.**"



NOTE

Failing to reject H_0 does **not** let you *accept* it. Maybe there really is no effect, or maybe there is one and your study just couldn't see it.

Don't worship the p-value

"Not significant" is not the same as "no effect." A real difference may be there; your study was just too small to see it.



Statistical $\neq$ practical

0.05 is a convention, not a hard cutoff. A real effect can sit just past it: $p = 0.06$ isn't proof of "no effect," so weigh borderline values with care.



The fix

Bigger samples, reported **effect sizes**, and a **power** analysis, all covered next.



IT'S A WIDESPREAD MISTAKE

Reviews of the published literature keep finding the same thing: a large share of papers wrongly treat "not statistically significant" as proof of "no effect." Don't join them.

Self-check

A trial reports $p = 0.07$ for a promising drug and concludes "the drug doesn't work." **What's wrong with that conclusion?**



ANSWER

$p = 0.07$ only means *we cannot say there is a difference* at $\alpha = 0.05$, **not** that the drug has no effect. The study may simply be **underpowered**.

The right move: report the **effect size** and **confidence interval**, and ask whether a larger sample would have detected a difference that may well be real (and practically meaningful).

• PART 8 • BEYOND P-VALUES

Effect size, power & confidence

A p-value says *whether*. These tools say *how much*, *how sure*, and *how big a study you need*.

Two mindsets

There are two distinct ways to ask a question of your data, and they emphasize different numbers.

| Hypothesis testing

"Does a difference occur or not?"

Emphasis on **statistical significance** and **p-values**. A yes/no verdict.

| Parameter estimation

"How big is the difference, and how sure are we?"

Emphasis on **95% confidence intervals** and **effect sizes**, often drawn from prior literature or a pilot study.

Effect size: how big is the difference?

Significance tells you an effect probably exists; **effect size** tells you whether it's worth caring about. It's the ratio of the change to the spread.

Cohen's d = $(\text{mean}_1 - \text{mean}_2) \div \text{pooled standard deviation}$



d ≈ 0.2

Small effect



d ≈ 0.5

Moderate effect



d ≈ 0.8

Large effect



TIP

For relationships rather than group means, **Pearson's r** plays the same role, capturing the strength and direction of a linear effect.

Effect size in action: how big is "d = 0.5"?

Two teaching methods are compared on a 100-point exam. Does the 5-point gap actually matter? Effect size turns that raw gap into something interpretable.

1 • The calculation

- **Method A** averages 75
- **Method B** averages 80
- Scores spread with **SD = 10** (both groups)

$$d = \frac{\text{mean}_B - \text{mean}_A}{\text{SD}} = \frac{80 - 75}{10} = 0.5$$

A **moderate** effect, real enough to notice, not merely "statistically present."

2 • What 0.5 means

The averages sit **half a standard deviation** apart, so a student at B's average (80) lands at **z = +0.5** on Method A's score curve.

$$\Phi(0.5) \approx 0.69$$

The average Method-B student **outscores ~69%** of Method-A students. (*This percentile is Cohen's U_3 .*)



WHY 69%?

Φ is the normal **cumulative** curve: about 69% of a normal group falls below a point sitting 0.5 SD above the mean. The reading assumes scores are roughly **normal with similar spread**; skew or very unequal variances make 69% only approximate.

A caution: two "superiority" numbers for one d

For the same $d = 0.5$, people quote two different percentages. They sound interchangeable but answer **different questions**, so don't mix them up.



Cohen's U_3

$$U_3 = \Phi(d) \approx 69\%$$

Where does B's *average* student rank within A? A student at B's mean sits at the **69th percentile** of group A.



Common Language Effect Size (CLES)

$$\text{CLES} = \Phi(d/\sqrt{2}) \approx 64\%$$

What's the chance a *random* B beats a *random* A?
Pick one student from each group; B scores higher
~64% of the time.



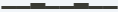
WHICH ONE THE EXAMPLE USES

The previous slide says "*the average Method-B student outscores ~69%*": that's U_3 , and the number is correct as written. Reframe it as "*a random B beats a random A*" and the figure must change to **~64% (CLES)**.

Signal vs. noise

A real effect (signal) has to stand out from natural variability (noise). The bigger the noise, the harder a given effect is to detect.

Low noise — signal pops

1 noise floor: 
2 signal:  ← easy to see

High noise — signal buried

1 noise floor: 
2 signal:  ← lost in the chatter



TIP

The lever you control is **sample size**: high variance or large standard deviation? Compensate with a **bigger n** so the signal can rise above the noise.

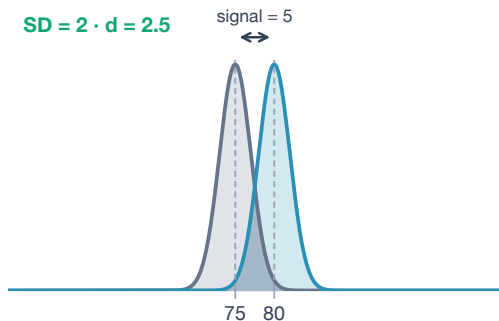
Signal ÷ noise = Cohen's d

Effect size isn't a new idea; it just **names the pieces** of that picture. Signal is the gap; noise is the spread; d is one over the other.

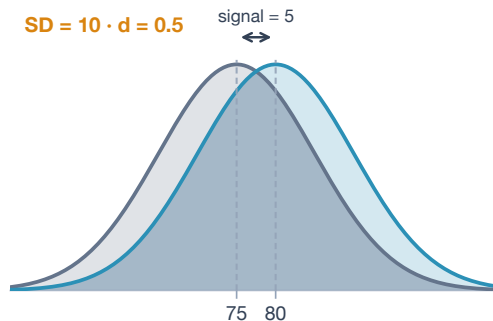
$$d = \frac{\text{signal}}{\text{noise}} = \frac{\text{mean}_B - \text{mean}_A}{\text{SD}_{\text{pooled}}}$$

Same skeleton as a t -statistic, $t = \frac{\text{difference}}{\text{standard error}}$: every "is this real?" test divides effect by spread.

Low noise — signal pops



High noise — signal buried



NOTE

Both panels carry the **same 5-point signal**; only the denominator changes. The buried one ($d = 0.5$) is exactly the exam example from earlier; shrink the noise and the **same gap** leaps out ($d = 2.5$).

Power & sample size

Statistical power is the probability of detecting an effect that's really there. It's the flip side of the false-negative rate:

Power = $1 - \beta$ (β is the Type II error rate)



What raises power

A **bigger sample**, a **larger true effect**, **less variance**, or a more lenient α .



What it buys you

Fewer **Type II errors**: fewer real effects missed.



TIP

Run a **power analysis** while designing the study: feed in your expected effect size and α , and it tells you the **sample size** needed to detect the effect. Do this *before* collecting data, not after.

The 95% confidence interval

Two different "95%" ranges get confused constantly, and the split is exactly **SD vs. SE**.

| 95% CI — precision of the **mean**

$\bar{x} \pm 1.96 \times SE$, where $SE = SD \div \sqrt{n}$

Across repeated samples, ~95% of such intervals contain the true mean. Crucially, it **narrows as n grows**, because more data pins the mean down.

| ± 1.96 SD — spread of the **data**

For a normal curve, ± 1.96 SD around the mean covers ~95% of **individual values** (the empirical rule). This range **does not shrink** with n.

- ± 1 SD $\approx 68\%$
- ± 2 SD $\approx 95.45\%$
- ± 3 SD $\approx 99.73\%$



SD IS NOT SE

Standard deviation describes the spread of the data; **standard error** ($SD \div \sqrt{n}$) describes the precision of the *mean*. The 95% CI uses **SE**, not SD, so label your error bars so readers know which one you plotted.

Two clinical thresholds worth knowing

When estimating effects, ask not "is it detectable?" but "is it *meaningful*?"



MCID

Minimally Clinically Important Difference: the smallest change a patient would actually *notice or value*. Statistical significance below this is academic.



MDC

Minimally Detectable Change: the smallest change your *measurement* can reliably distinguish from noise.



TIP

A well-designed study reports both: it can **detect** a change (MDC) and that change is large enough to **matter** (MCID).

Self-check

Two studies test the same drug. Both report $p = 0.04$. Study A has $d = 0.2$; Study B has $d = 0.8$. Are they equally important?



ANSWER

No. Both cleared significance, but **effect size tells the real story**: Study B's effect ($d = 0.8$, large) is four times the magnitude of Study A's ($d = 0.2$, small).

A tiny p-value with a tiny effect often just means a **big sample**: statistically detectable, but maybe below the **MCID** and not clinically meaningful.

■ PART 9 · STUDY DESIGN

Design the study, not just the test

The best statistics can't rescue a badly designed study. Plan the analysis *into* the design from day one.

The design checklist

Decide these *before* collecting data, since they all shape which analysis is possible.

Sampling

- Expected **sample size**
- **Randomization** procedure
- Single- or **double-blind**
- Inclusion / **exclusion** criteria

Comparison

- A **control group** for meaningful contrast
- The **statistical test** you intend to use
- Planned tests for **normality** & equal variance

Hazards

- **Missing data** & unequal group sizes
- **Confounding** variables
- **Ethics** & data-safety monitoring
- A **data-management** scheme

PICOT / PECOT

A framework that forces a complete research question. **PICOT** (intervention) and **PECOT** (exposure) share the same five slots.



P

Population studied



I / E

Intervention or
Exposure



C

Comparison group



O

Outcome measured



T

Time / duration



NOTE

I = something you *do* to the group (a treatment); **E** = something they're *exposed* to (e.g. a risk factor).
Observational and qualitative studies may not fill every slot.

Weak vs. strong objective

A good study objective follows *logically* from a complete PICOT/PECOT. Compare two real-world framings.

Needs work

"Is cannabis use in adolescents associated with later depression, anxiety, or suicidal behavior?"

- P adolescents ✓
- E cannabis ✓
- C *missing* (no non-user group)
- O depression / anxiety / suicidality ✓
- T *missing* (no time frame)

Well-defined

"Does 12 months of a remote aerobic-exercise program prevent memory loss in elderly people with MCI, vs. memory training alone?"

- P elderly with mild cognitive impairment ✓
- I aerobic exercise ✓
- C memory-training group ✓
- O memory function (MMSE) ✓
- T 12-month program, 1-year follow-up ✓

Even observational studies fit the frame

PECOT works without an intervention: you just *stratify* by exposure and track over time.



EXAMPLE — DEMENTIA & WALKING AFTER STROKE

Question: Does the presence of dementia affect walking ability one year after a stroke?

- **P** stroke patients · **E** dementia present vs. absent · **C** the two strata
- **O** walking speed; relationship to cognition (MMSE) via linear regression
- **T** baseline vs. **one year** later



TIP

The **T** is what makes it a study and not a snapshot: measuring the *same* patients at baseline and one year reveals *change*, not just a single difference.

Accuracy vs. precision

Two different virtues. **Accuracy** = on target (calibrated). **Precision** = repeatable (tight cluster). You want both.



High accuracy · high precision ✓

1	· ◎ ·	tight cluster,
2	◎◎	on the bullseye
3	· ◎ ·	

The goal of every study.



High precision · low accuracy

1	◎◎ ·	tight cluster,
2	◎◎	off-centre
3	· · ·	

Often an **uncalibrated** instrument.



Low precision · high accuracy

1	· ◎ · · ◎	scattered, but
2	· ◎ ·	centred on target
3	◎ · · ◎	

Noisy, yet unbiased on average.



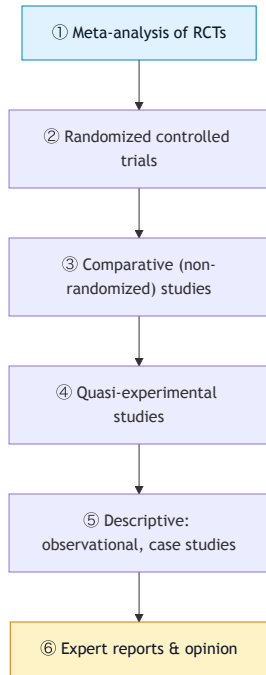
Neither

1	◎ · · · ·	
2	· · · ◎	
3	· · ◎ · ·	

Scattered **and** off-target.

The hierarchy of evidence

The study design itself determines how much **bias** the result can carry. Less bias → stronger evidence.



⚠ Top = least bias, highest quality. A single case study ($n = 1$) sits near the bottom, with high potential for bias.

Reducing bias and error

You can't eliminate bias, but design choices shrink it. Anticipate the analysis and you reduce the influence of chance *and* bias.



Randomize

Random assignment breaks the link between treatment and hidden differences.



Increase sample size

More data dilutes random chance and raises power.



Strict criteria

Clear inclusion / exclusion rules keep the sample comparable.



Analyze subgroups

Measure and model the factors that might bias results.



TIP

Best of all, where ethical and feasible: a **randomized controlled trial**, or a meta-analysis of several.

Self-check

A colleague's objective reads: "*We studied a new inhaler in asthma patients and measured lung function.*" **Which PICOT slots are missing, and why does it matter?**



ANSWER

Missing the **Comparison** (no control / standard-inhaler group) and the **Time** frame.

Without a comparison group you can't attribute any change to the inhaler rather than to time or placebo; without a defined follow-up you don't know *when* lung function was measured. Both gaps weaken the evidence and leave the analysis underspecified.

■ PART 10 · TOOLS & REPORTING

Show the work

Once the analysis is done, the result has to be *seen*. Good visualization, in the right file format, is part of the science.

After the test: visualize

A p-value in prose is forgettable; a clear figure is convincing. Plot your results as **graphs, figures, and tables**.



Figures

Scatterplots, box/violin plots, regression lines: the shape of the result at a glance.



Tables

Exact values: means, p-values, confidence intervals, sample sizes.



TIP

Every researcher has artwork needs (figures, schematics, illustrations) for manuscripts, posters, and talks. Treat data visualization as a first-class skill, not an afterthought.

Vector vs. raster

The format you save figures in decides whether they survive editing and zooming. For journals, this matters a lot.

Vector — use this

- **Mathematically described** lines & points
- Edit and **resize with no quality loss**
- Stays crisp at any zoom
- PDF / SVG / EPS · Illustrator · PowerPoint

Journals prefer vector for plots and diagrams.

Raster — avoid for plots

- A grid of **pixels** (a bitmap)
- **Not editable** point-by-point
- **Blurs and pixelates** when zoomed
- JPEG / PNG · scanned images

Fine for photographs; poor for charts.

■ PART 11 · REVIEW & CHEAT-SHEET

Putting it all together

The whole deck, distilled: take-home messages, a which-test table, and a one-screen reference.

Take-home messages



Match test to data

Choose the test by the **type and distribution** of your data; explore it first.



Check assumptions

Test **normality** and **equal variance**; know which **parameters** you'll estimate.



Ask a clear question

A clean **PICOT/PECOT**: control group, time element, objective the test fits.



Design out bias

Raise **accuracy & precision**, lower bias, ideally with a **randomized controlled trial**.

Which test do I use?

The lookup table to keep beside you. The two questions that drive it: **how many groups**, and **is the data normal?**

Goal	Normal / continuous	Non-normal / ordinal
Compare 2 groups	Independent t-test	Wilcoxon–Mann–Whitney
Compare 3+ groups	One-way ANOVA → Tukey	Kruskal–Wallis → Dunn
Measure a relationship	Pearson's r / regression	Spearman's rho
Compare counts / categories	—	Chi-square

⚠ Running many pairwise t-tests instead of an ANOVA? Apply a **Bonferroni** correction to the p-values.

The workflow on one screen

1 · Explore

- Name each variable's **scale**
- Frequency table + **histogram**
- Box / **violin** plot for shape
- Median for skew & outliers

2 · Check & choose

- Normality → **Shapiro–Wilk**
- Variance → **Levene / Bartlett**
- Not normal → **non-parametric**
- Variance only → **Welch** (parametric)
- Pick test by **#groups**

3 · Test & report

- State H_0 / H_1 , set $\alpha = 0.05$
- Report the **p-value** *and*
- **effect size** (Cohen's d)
- **95% CI**, sample size, power

4 · Design (do first!)

- **PICOT / PECOT** question
- **Control** group + **time**
- **Randomize**; set criteria
- Power analysis → **sample size**

Glossary, in one breath

p-value if there were truly no effect, the chance of a result this strong or stronger by luck alone

α pre-set significance threshold (usually 0.05)

Type I false positive: reject a true H_0

95% CI plausible range for the mean: $\bar{x} \pm 1.96 \text{ SE}$ ($\text{SE} = \text{SD} / \sqrt{n}$)

Homoscedasticity equal variances across groups

Effect size the difference's size relative to the data's spread (Cohen's d): whether it matters, not just that it exists

Power $1 - \beta$; chance of catching a real effect

Type II false negative: miss a real effect (β)

SD vs SE spread of data vs precision of the mean

Post-hoc which groups differ (Tukey / Dunn)

Advanced tips to carry forward



SIGNIFICANT $\neq$ IMPORTANT

A tiny p-value with a tiny effect size is often just a large sample. Always read **effect size** and **CI** alongside p.



ABSENCE OF EVIDENCE

"Not significant" means *we cannot say there's a difference, never there is none*. Beware the underpowered null.



PRE-SPECIFY EVERYTHING

Fix your α , **hypotheses**, and **analysis plan** **before** seeing data. Post-hoc choices inflate false positives.



CORRELATION $\neq$ CAUSATION

A strong r shows association, not cause. Interpret regression with the p-value *and* domain judgement.

You can choose — and defend — a test now

Read the data's shape → check the assumptions → pick the matching test → report the effect, not just the p-value → and design the study so the answer is trustworthy.

Press for the overview; this deck is yours to revisit as a reference.